The biregular case: supplementary material to the weighted plane class and its obstructions

Vico Bonfioli*

August 2026

Abstract

This note carries the material removed from [33] when that paper was cut to its spine: the counting refinement and its barriers, the expected non-backtracking polynomial, the saturation measurements, the coordinate case at rank b with the sharpness of Xus bound, the account of what a proof would have to contain, and the full formalization record. Nothing here is needed to read [33].

1 A counting refinement, and its two barriers

2 A counting refinement

GAPCOUNT is refuted along with the conjecture it refines, and this section is kept because most of it does not depend on GAPCOUNT being true. What survives, and is what the section is now for: the implication GAPCOUNT ⇒ localization and the boundary conditions, machine-checked; BAND, which is a theorem about the abelian shadow and independent of either conjecture; the settled cases at feedback vertex number one and, conditionally, two; and the two barriers, which close routes that would otherwise look open. The sweeps are retained as a record of where the failure is *not* visible. A reader interested only in the biregular case may go straight to §2 and the operator form.

Conjecture 1 asserts a non-vanishing, which offers no handle. This section records a strictly stronger statement that equates two integers, the cases it settles, and the two barriers that close off the obvious routes to the rest. Labels are stated throughout: what is machine-checked, what is proved modulo a cited classical theorem, and what is open.

Let T be the universal cover of a finite connected loopless G on n vertices, and $b = b_1(G)$. Writing A_G for the universal-cover adjacency operator as a matrix over $C_r^*(F_b)$, a real x lies outside $\operatorname{spec}(T)$ exactly when $xI - A_G$ is invertible there, and then the trace of its negative spectral projection,

$$\kappa(x) = \tau_n\big(\mathbf{1}_{(-\infty,0)}(xI - A_G)\big),$$

is an integer in $[0, n]$, because $K_0(C_r^*(F_b)) = \mathbb{Z} \cdot [1]$ by Pimsner–Voiculescu.

Conjecture 1 (GAPCOUNT)**.** *For every $x \notin \operatorname{spec}(T)$, the number of roots of μ_G strictly above x, with multiplicity, equals $\kappa(x)$.*

*Lean 4 sources: `RamaLean/` in the `rama-notes` repository. Correspondence: `vicobonfioli@gmail.com`.

Conjecture 1 implies Conjecture 1 at $d = 1$: κ is constant across a gap while the root count jumps at any root, so a gap contains none. By the counterexample of [27] the implication now runs the other way and Conjecture 1 is false as well, on the same graph: the root $\sqrt{5}$ crosses inside a gap on which κ is constant. We state it as it was, because the implication and the boundary conditions are what the sweeps below actually test, and because the counting refinement remains the right object for whichever class of graphs turns out to satisfy the localization. That implication, the fact that an integer-valued locally constant function is determined on a connected gap, and the boundary conditions $\kappa = 0$ above $\rho(T)$ and $\kappa = n$ below $-\rho(T)$, are machine-checked in `RamaLean/GapLabel.lean`. All the content is in what happens as a band is crossed.

2.1 The abelian shadow

Put a circle variable on each cotree edge and let $A_G(z)$ be the magnetic adjacency matrix, $B_k = \lambda_k(\mathbb{T}^b)$ the k-th Floquet band, and $\theta_1 \geq \cdots \geq \theta_n$ the roots of μ_G.

Proposition 2 (BAND). *$\theta_k \in B_k$ for every k. Consequently, for $x \notin \mathrm{spec}(G^{\mathrm{ab}})$ the number of roots above x equals the number of bands above x.*

Proof. Godsil–Gutman writes μ_G as the average characteristic polynomial over ± 1 edge signings, and Marcus–Spielman–Srivastava [23] make those signings an interlacing family. The common interlacing lemma brackets the average at every index, not only the largest, so $\lambda_k(A_s) \leq \theta_k \leq \lambda_k(A_{s'})$ for two signings. Both are points of $\mathbb{T}^b$, and B_k is the continuous image of a connected space, hence an interval. □

Everything downstream of those two classical inputs is machine-checked in `RamaLean/BandTheorem.lean`: that bands are order connected (`ordConnected_range_of_continuous`), the squeeze (`mem_of_squeezed`), the count identity (`countAbove_eq_countBands`) and the recovery of the unicyclic case (`conj10_of_bands_subset`). The inputs themselves enter `band_of` as hypotheses.

The core of the common interlacing lemma is machine-checked in `RamaLean/Interlacing.lean`: a sign argument fixes the parity of a root count and nothing more, an interlacing confines that count to a window of width two, and parity with such a window determines the value. What is not formalized, and is cited to [23], is that the signings have a common interlacing at all.

Proposition 2 settles Conjecture 1 whenever every band lies in $\mathrm{spec}(T)$, which is exactly $b_1(G) = 1$, since $F_1 = \mathbb{Z}$ makes the maximal abelian cover the universal cover. That recovers [34] by a different route and sharpens it to $\theta_k \in B_k$. The hypothesis fails at $b_1 \geq 2$: for K_4 the top band carries the Perron value 3 while $\rho(T) = 2\sqrt{2}$.

The counting half of Proposition 2 does not need [23]. If x avoids every $\lambda_k(z)$ then $\{z : \lambda_k(z) > x\}$ is clopen, so on a connected parameter space the eigenvalue count above x is the same integer for every z; every member of the family then has the same sign at x, so the average does too. Parity follows at once, and parity upgrades to equality by counting: crossing one component of the spectrum, each component carries an odd number of roots, and n odd positive integers summing to n are all 1. Both steps are machine-checked in `RamaLean/BandCount.lean`.

2.2 Feedback vertex number one

Suppose some vertex v has $F = G - v$ a forest, so every cycle passes through v; the first Betti number is unrestricted, and flowers, theta graphs and every $K_{2,q}$ qualify.

Proposition 3. *Conjecture 1, hence Conjecture 1 at $d = 1$, holds for every such G, modulo Pimsner–Voiculescu and Godsil's path tree.*

Proof. Eliminating the preimage of F gives a Schur complement $S_v(x)$ which is a *finite* sum of group elements, so it lies in the group algebra inside $C_r^*(F_b)$ and not merely in $L(F_b)$; that distinction is what makes projectionlessness applicable, since $L(F_b)$ is a II_1 factor full of projections. Haynsworth additivity gives $\kappa(x) = N_F(x) + \delta(x)$ with $\delta(x) \in \{0, 1\}$, and $\tau(S_v(x)) = \mu_G(x)/\mu_{G-v}(x)$, so definiteness fixes the parity of $N_G - N_F$. Godsil's path tree makes μ_{G-v} interlace μ_G, confining $N_G - N_F$ to $\{0, 1\}$. A window of width two and a parity determine the value. □

The assembly is `FeedbackGapCount.gapcount_fvs_one_of`, with the three classical inputs as explicit hypotheses, and `conj10_fvs_one_of` composes it with the skeleton of Section 2.

Three remarks on that proof. The combinatorial core, that in an acyclic graph two distinct neighbours of v cannot be joined without passing through v, is machine-checked (`FeedbackVertex.no_bypass`); it is why a walk returning to the same lift must enter and leave at the same attachment point. The trace formula requires G *simple*: the argument uses that a lifted attachment point has exactly one neighbouring lift of v. Haynsworth additivity needs no appeal to Sylvester's law in a von Neumann algebra: the block matrix is congruent to the direct sum of the corner and the Schur complement by explicit unipotent factors, and scaling the off-diagonal part keeps every point of the path invertible, so the trace of the negative spectral projection is constant along it. That is machine-checked in `RamaLean/Congruence.lean`.

2.3 Feedback vertex number two, and where it stops

At $k = 2$ the argument breaks in a locatable place: $M_2(C_r^*(F_b))$ is *not* projectionless, $\mathrm{diag}(1, 0)$ being a counterexample, so $S(x)$ can be indefinite. What survives is the K-theory, which keeps $\delta(x) \in \{0, 1, 2\}$. Two deletions give a window of width three, so parity leaves a two-way choice between $\delta = 0$ and $\delta = 2$, and one further bit is needed. That bit exists.

Proposition 4. *Let $F = G - v_1 - v_2$ and let a, b_1, b_2, c count the roots above x of $\mu_G, \mu_{G-v_1}, \mu_{G-v_2}, \mu_F$. If $a - c$ is even then $b_1 = b_2$, and*

$$a = c \iff \mu_F(x) \text{ and } \mu_{G-v_1}(x) + \mu_{G-v_2}(x) \text{ agree in sign.}$$

Proof. Vertex-deletion interlacing gives $a - b_i \in \{0, 1\}$ and $b_i - c \in \{0, 1\}$. If $a = c$ then $c \le b_i \le a = c$; if $a = c + 2$ then $c + 1 = a - 1 \le b_i \le c + 1$. Either way $b_1 = b_2$, so the two polynomials share a sign at x and so does their sum. □

Proposition 4 is machine-checked in `RamaLean/TieBreak.lean`, and it needs no hypothesis about $\mathrm{spec}(T)$. It is not enough. The remaining gap is the parity itself: at $k = 1$ it came free from $\tau(S_v) = \mu_G/\mu_F$ and definiteness, while at $k = 2$ the corresponding object is the constant term of the abelianised determinant of a 2×2 matrix over a noncommutative algebra, whose sign has no evident reason to be $(-1)^\delta$. So feedback vertex number two is *reduced, not proved*. The arithmetic that would close it once the parity is supplied, a window of width three together with a parity and one tie-break bit, is `FeedbackTwo.eq_of_window3`, and `gapcount_fvs_two_sign` discharges the tie-break using Proposition 4, leaving the parity as the sole remaining hypothesis.

What it reduces to is now quantitative. Since μ_G/μ_F is the average over $\mathbb{T}^b$ of $\det S(x,z)$, and $\operatorname{sign}\det S(x,z) = (-1)^{\delta_{\mathrm{ab}}(x,z)}$, writing I_j for the integral of $|\det S|$ over $\{\delta_{\mathrm{ab}} = j\}$ gives $\mu_G/\mu_F = I_0 - I_1 + I_2$, and the required sign is exactly the assertion that the parity class matching δ dominates the others *in integral*. Measurement (`code/inertia_split.py`) shows the split is lopsided but not degenerate: at two triangles joined by an edge, in a genuine gap with $\delta = 1$, the class $\delta_{\mathrm{ab}} = 1$ occupies 0.987 of the torus and $\delta_{\mathrm{ab}} = 2$ occupies 0.013. The integrand therefore does change sign, so no argument asserting that it does not can work. The decomposition and the criterion are recorded in `RamaLean/InertiaSplit.lean`, with a witness that measure alone does not decide it.

2.4 Two barriers

Both close off routes that look available.

Proposition 5. *For $b \geq 2$ there is no unital $*$-homomorphism $C^*_r(F_b) \to C(\mathbb{T}^b)$ carrying the generators to the coordinates.*

Proof. Such a map would carry units to units, so $S(x)$ invertible would force $S(x,z)$ invertible for every z, hence $\det(xI - A_G(z)) = \mu_F(x)\det S(x,z)$ nonzero for every z, hence $\operatorname{spec}(G^{\mathrm{ab}}) \subseteq \operatorname{spec}(T) \cup \{\mu_F = 0\}$. For K_4 with a two-element feedback set that fails at $x = 3$: the Perron value lies in $\operatorname{spec}(G^{\mathrm{ab}})$, while $\rho(T) = 2\sqrt{2} < 3$ and $\mu_{K_2}(3) = 8$. □

The refutation shape is machine-checked in `RamaLean/NoQuotient.lean`. The underlying reason is that a quotient descends to reduced C^*-algebras when the *kernel* is amenable, and $[F_b, F_b]$ is free of infinite rank.

The second barrier is narrower than it first appeared. Every argument that runs on the torus family needs x outside $\operatorname{spec}(G^{\mathrm{ab}})$, and on the smallest examples that fails: for K_4, two triangles joined by an edge, the theta graph and K_4 with pendants, the internal gaps of $\operatorname{spec}(T)$ are swallowed by bands of the abelian cover. It would be wrong to read that as a general obstruction. A certified search over two growing families (`code/torus_gg.py`) finds the opposite behaviour as soon as the graphs are larger: ten of fourteen gap midpoints lie outside $\operatorname{spec}(G^{\mathrm{ab}})$, with a margin that grows with the number of vertices: for two c-cycles joined by an edge it runs 1.18, 7.52, 21.50, 51.35 as c goes from four to seven. So the torus family is vacuous on the smallest graphs and informative as they grow, the reverse of what the first computation suggested, and the localization of [34] settles Conjecture 1 outright at every such point, for every first Betti number and with no feedback vertex hypothesis.

Nor is that settlement subsumed by Heilmann–Lieb. Those graphs have maximum degree three, so Heilmann–Lieb excludes roots only outside $[-2\sqrt{2}, 2\sqrt{2}]$, while the certified points sit at $\pm 1.93, \pm 1.99, \pm 2.04, \pm 2.06$, all interior. The certification cannot be a grid minimum, which proves nothing when the zero set of $\det(xI - A_G(z))$ is a curve: that determinant has degree one in each z_e, so its 3^{b_1} Fourier coefficients are recovered exactly by a three-point transform, they bound the gradient, and a grid search together with that Lipschitz constant certifies a positive lower bound, while a sign change on the grid certifies the opposite verdict.

2.5 Evidence

`code/universal_cover.py` computes $\operatorname{spec}(T)$ and κ for an arbitrary finite graph, by the non-backtracking cavity equations on the directed edges. Its regression check is K_4 with a pendant at each vertex, where $\operatorname{spec}(T) = \pm[\sqrt{3}-\sqrt{2}, \sqrt{3}+\sqrt{2}]$ in closed form; the solver returns $[-3.1456, -0.3163]$ and $[0.3184, 3.1477]$. Runs whose density fails to integrate to one are discarded, since point spectrum makes the cavity fixed point singular and a diverged κ reads exactly like a violated Conjecture 1. That the cavity map preserves the lower half plane, so the iteration is total and the density non-negative, is machine-checked in `RamaLean/Cavity.lean`; convergence is not claimed.

A Rust sweep (`code/covercheck`) applies this to every connected graph of feedback vertex number at most two on up to seven vertices, computing the root count by Budan–Fourier, which is exact for a real-rooted polynomial, and κ by the cavity equations. Over 55 131 graphs and 99 487 gap points there is no failure of Conjecture 1 and none of Conjecture 1, no run is discarded by the validity gate, and κ is never further than 0.205 from an integer. That range could not have found the counterexample and should not be read as having looked: Hall's graph has 41 vertices and feedback vertex number five, so it lies outside the sweep in both parameters. What the sweep establishes is that the failure is not visible at small size, which is why it went unnoticed, and not that it is rare.

The measurement also shows what an estimate would have to exploit. The wrong-parity class is not merely small in measure, it is suppressed quadratically in integral: at two triangles joined by an edge, in the gap at $x = -1.767$, it occupies 0.013 of the torus and contributes 0.0007 of the integral of $|\det S|$, a margin of 0.9987. The reason is structural: the wrong-parity region is a shell around $\{z : \det S(x, z) = 0\}$, where one eigenvalue has just crossed and $|\det S|$ is correspondingly small. Small measure and small integrand together, which is why the competition is not close.

Scanning a whole gap rather than one point confirms the exponent. Across both internal gaps of two triangles joined by an edge, $I_{\text{wrong}}/m^2_{\text{wrong}}$ stays between 2.4 and 5.2 and the margin never falls below 0.977 (`code/margin_scan.py`); there is no blow-up at the gap edges, where the wrong-parity class is largest and the ratio is smallest. The chain that would turn this into an estimate is machine-checked in `RamaLean/ShellBound.lean`. Its first step is an identity rather than a bound: for a 2×2 Hermitian matrix the operator norm is the larger eigenvalue in absolute value, so $|\det S| = \|S\| \cdot \min(|\lambda_1|, |\lambda_2|)$ (`abs_mul_eq_max_mul_min`), and on the shell only the crossing eigenvalue matters. Weyl makes that eigenvalue Lipschitz in the phase, giving $|\det S(z)| \le ML \operatorname{dist}(z, \partial)$ and hence $I_{\text{wrong}} \le m_{\text{wrong}}\, ML\, \text{reach}$, where $\text{reach} = \sup_{z\in A} \operatorname{dist}(z, \partial A)$ is the inradius (`det_le_on_shell`). The relevant quantity is the inradius and not the diameter: a shell wrapping the torus has diameter of order one however thin it is.

The inradius is then bounded for free. A region of inradius r contains a ball of radius r, so $c_b r^b \le m$ and $\text{reach} \le (m/c_b)^{1/b}$, giving

$$I_{\text{wrong}} \;\le\; ML\, c_b^{-1/b}\, m_{\text{wrong}}^{1+1/b},$$

which at $b = 2$ is $m^{3/2}$ (`reach_le_sqrt`, `integral_le_three_halves`). The upper half of the estimate is therefore unconditional. The region is not a thin shell but a blob, with m of order reach^2: measurement gives reach/m running 2.60, 3.10, 4.63, 5.94, 9.80 as m falls from

0.052 to 0.0053. The exponent $3/2$ is accordingly weaker than the m^2 the numerics suggest, and stronger than anything needed. The lower half, a bound $I_{\text{right}} \geq c > 0$, is open, and its natural route is precisely what Proposition 5 rules out.

The obvious substitute is circular, and it is worth saying why. Splitting the average by inertia class gives $\mu_G/\mu_F = I_0 - I_1 + I_2$ *identically*, so $I_{\text{right}} - I_{\text{wrong}} = |\mu_G/\mu_F|$ and bounding I_{right} below by $|\mu_G/\mu_F|$ assumes exactly the sign at issue. No independent lower bound on $|\mu_G(x)|$ is available inside a gap, since Heilmann–Lieb gives root-free intervals only outside $[-\rho, \rho]$, where the conjecture is already known.

A substitute that is not circular is the geometric mean. Jensen gives

$$I_{\text{right}} + I_{\text{wrong}} \;=\; \int_{\mathbb{T}^b} |\det S| \;\geq\; \exp\Bigl(\int_{\mathbb{T}^b} \log|\det S|\Bigr) \;=:\; \Delta,$$

a Mahler measure of the abelian cover, which owes nothing to the sign of μ_G. Hence $I_{\text{right}} \geq \Delta - I_{\text{wrong}}$, and domination holds as soon as $\Delta > 2I_{\text{wrong}}$. Combined with the unconditional upper half this reduces feedback vertex number two to a single inequality, $\Delta > 2MLc^{-1/2}m^{3/2}$ (`RamaLean/JensenRoute.lean`). The geometric mean, which appeared above as the source of the obstruction, is here the tool.

Measured across both internal gaps of two triangles joined by an edge, Δ lies between 0.759 and 0.823 while I_{wrong} runs from 10^{-4} to 10^{-2}, so $\Delta/2I_{\text{wrong}}$ runs between 41 and 3927 (`code/jensen_route.py`); the same script verifies the circularity identity to 10^{-15}. A sweep over 1475 gap points of graphs of feedback vertex number at most two with $2 \leq b \leq 4$ records no failure of $\Delta > 2I_{\text{wrong}}$ and a worst ratio of 298, and the ratio does not degrade with b: its minimum is 298 at $b = 2$, 2070 at $b = 3$ and 1740 at $b = 4$ (`code/jensen_sweep.py`). What is a Mahler measure over $|\mu_F(x)|$ is recorded in `RamaLean/MahlerRoute.lean`.

2.6 Why the feedback vertex set should be removed

The inequality $\Delta > 2MLc^{-1/2}m^{3/2}$ is not the right target, and the reason is an artefact of the Schur complement rather than a feature of the conjecture. Both $M = \sup\|S\|$ and the Lipschitz constant L scale like $1/\mu_F(x)$, while Δ scales like $1/\mu_F(x)$; so as x approaches a root of μ_F inside a gap the left side grows one whole factor of μ_F faster than the right, and the inequality fails at points where nothing is wrong. Roots of μ_F are not excluded from gaps of $\operatorname{spec}(T)$.

The Schur complement is a computational device and can be dropped. With $P_x(z) = \det(xI - A_G(z)) = \prod_k (x - \lambda_k(z))$ and $N(z) = \#\{k : \lambda_k(z) > x\}$, the sign of P_x is $(-1)^{N(z)}$, and the localization of [34] gives

$$\mu_G(x) = \int_{\mathbb{T}^b} P_x = I_{\text{even}} - I_{\text{odd}}, \qquad I_p = \int_{\{N \equiv p \ (2)\}} |P_x|,$$

with no feedback vertex hypothesis, no restriction on b, and no μ_F. It also makes the Lipschitz constant explicit instead of measured, and better than the obvious bound. Crudely, $\partial A/\partial\theta_j$ has a single entry of modulus one together with its conjugate, so its operator norm is 1 and $\|A(z) - A(w)\| \leq \sum_j |\Delta\theta_j|$, which costs a factor b. First-order perturbation theory does better: $\partial\lambda_k/\partial\theta_j$ is the quadratic form of that one edge against a unit eigenvector v, so it is at most $|v_{u_j}|^2 + |v_{v_j}|^2$, and summing over cotree edges charges each vertex once per incident cotree edge. Hence

$$\sum_j \left|\frac{\partial \lambda_k}{\partial \theta_j}\right| \;\le\; D,$$

with D the largest number of cotree edges at a vertex. $D \le 2$ whenever the cotree edges can be chosen vertex-disjoint and $D \le \Delta(G)$ always, so D does not grow with the graph while b does. The per-edge estimate (`BandLipschitz.quad_form_bound`), the double count (`sum_deg_bound`) and the assembly (`grad_l1_bound`) are machine-checked; first-order perturbation theory is not in Mathlib and enters `grad_l1_bound` as a hypothesis in the shape the argument consumes.

Let κ be the number of bands whose range contains x. If $\kappa = 0$ the localization settles x. Over 149 764 residue points from 40 146 graphs on up to eight vertices, $|K| = 1$ at 140 661 and $|K| = 2$ at 9103, and no point has $|K| \ge 3$ (`code/jsweep`). How often $|K| = 0$ happens depends sharply on b: 80.2% of 486 midpoints with $b = 2$ lie outside $\mathrm{spec}(G^{\mathrm{ab}})$, against 3.4% of 493 midpoints with $b = 3$, where a further 69.8% are certified inside and 26.8% undecided, so at most 30% of the $b = 3$ points can be settled this way (`code/residue_sweep.py`, `code/residue_b3.py`). More torus directions fill the gaps in. Over 5971 residue points, with $\kappa = 1$ at 5397 and $\kappa = 2$ at 574 and no point with $\kappa \ge 3$, the minority-parity region has measure at most 0.0674 and $\min(I_{\mathrm{even}}, I_{\mathrm{odd}})/\max(I_{\mathrm{even}}, I_{\mathrm{odd}})$ is at most 0.0169 (`code/minority.py`), so the two classes are not close to cancelling.

Proposition 6. *Suppose exactly one band function λ_{k_0} attains x on $\mathbb{T}^b$. Then $Q = \prod_{k \ne k_0}(x - \lambda_k)$ has constant sign s, and*

$$\mu_G(x) = s \int_{\mathbb{T}^b} \big(x - \lambda_{k_0}(z)\big)\, |Q(z)|\, dz.$$

Consequently $\mu_G(x) = 0$ if and only if x is the $|Q|$-weighted mean of λ_{k_0}.

Proof. The factors other than k_0 never vanish and $\mathbb{T}^b$ is connected, so Q has one sign. The integral is affine in x, so it vanishes exactly when $x \int |Q| = \int \lambda_{k_0} |Q|$. $\square$

Machine-checked as `ParitySplit.zero_iff_weighted_mean` and `ne_zero_iff_ne_weighted_mean`, with the identity verified to 10^{-15} in `code/kap.py`. A weighted mean lies between the bounds of the function averaged (`mean_mem_Icc`), so an x outside the range of the crossing band cannot be that mean: `ne_zero_of_outside_band` recovers the localization in this case with no estimate at all. Conjecture 1 at a point with $\kappa = 1$ becomes the statement that the $|Q|$-weighted mean of the crossing band lies in $\mathrm{spec}(T)$. Measured means are $-2.149, -1.483, 1.483, 2.143, 2.150$ on gaps whose interiors sit near ± 1.77, so they are far from x and inside $\mathrm{spec}(T)$.

Two caveats keep this in proportion. First, over 4900 gap midpoints $\kappa = 1$ at 2206 residue points and $\kappa \ge 2$ at 183, so Proposition 6 covers 92.3% of the residue and not all of it. Second, the statement it reduces to,

$$\text{the } |Q|\text{-weighted mean of the crossing band lies in } \mathrm{spec}(T),$$

is a conjecture and is labelled as one; what can be said is that a deliberate attempt to break it failed. Over all 2206 points with $\kappa = 1$ the weighted mean lies in $\mathrm{spec}(T)$ every time and is never closer than 0.167 to x (`code/wmean_test.py`), and a single coincidence there would have refuted Conjecture 1 outright.

2.7 Separating the band geometry from the weight

Proposition 6 mixes two unrelated things: where the crossing band sits relative to x, and how much the remaining factors vary. Separating them removes the restriction to one crossing band and gives a criterion whose two halves can be attacked independently.

Let $K = \{k : x \in B_k\}$ be the crossing set and put

$$\pi(z) = \prod_{k \in K} \big(x - \lambda_k(z)\big), \qquad w(z) = \Big|\prod_{k \notin K} \big(x - \lambda_k(z)\big)\Big|.$$

The factors outside K never vanish, so their product has a constant sign s and $\mu_G(x) = s \int \pi w$.

Proposition 7. *Write $\pi = \pi_+ - \pi_-$ and $J_\pm = \int_{\mathbb{T}^b} \pi_\pm$, and suppose $0 < q_{\mathrm{lo}} \le w \le q_{\mathrm{hi}}$. If*

$$\frac{J_+}{J_-} \;>\; \Lambda := \frac{q_{\mathrm{hi}}}{q_{\mathrm{lo}}},$$

then $\mu_G(x) \neq 0$. No hypothesis on $|K|$ is used.

Proof. $\int \pi w = \int \pi_+ w - \int \pi_- w \ge q_{\mathrm{lo}} J_+ - q_{\mathrm{hi}} J_- > 0$. □

Machine-checked as `CrossingSplit.ne_zero_of_split`, with the ratio form as `crit_of_ratio`. The point is that $J_\pm$ involve only the crossing bands and Λ only the rest, so a bound on either half is useful on its own; the weighted mean of Proposition 6 admits no such division. It is also not the wasteful estimate: bounding the whole integrand by $\sup w$ loses the four orders of magnitude that $\sup w / \inf w$ can reach, whereas here $\inf w$ is applied to the majority term and $\sup w$ only to the minority term.

The criterion is not always satisfied. Over 149 764 residue points drawn from 40 146 graphs on up to eight vertices it holds 149 511 times and fails 253 times, with worst margin 0.0616 (`code/jsweep`). A sample of 208 points had shown it holding every time, which was too small to see the failures.

Two facts sharpen what is left. Every failure has exactly one crossing band: at $|K| = 2$ the criterion holds at all 9103 points, and no point with $|K| \ge 3$ occurs in the corpus at all. And Conjecture 1 is untouched at the failing points, the true ratio $\min(I_+, I_-)/\max(I_+, I_-)$ never exceeding 0.0233 over the whole corpus, so the conclusion holds comfortably where this sufficient condition does not see it. The worst case is $n = 8$, $b = 3$, $|K| = 1$, $x = 2.0607$ on the graph with edges $01, 02, 03, 04, 14, 25, 36, 37, 56, 57$.

What the criterion buys is therefore a reduction covering 99.83% of the residue, together with a target that has structure: J_+/J_- is large because x sits near an edge of the crossing band, the minority measure never exceeding 0.0674, and Λ is a product of band-distance ratios over the non-crossing bands. Bounding either one is a self-contained problem, and the 253 exceptions are a small explicit list to be understood rather than a diffuse obstruction.

The weight can be removed from the criterion entirely. A product is bounded above by the product of the upper bounds of its factors and below by the product of the lower bounds, and for a band that does not cross x the range $B_k = [\ell_k, h_k]$ lies on one side, so the corresponding factor is exactly $1 + \mathrm{width}(B_k)/\operatorname{dist}(x, B_k)$. Hence

$$\Lambda \;\le\; \prod_{k \notin K} \Big(1 + \frac{\mathrm{width}(B_k)}{\operatorname{dist}(x, B_k)}\Big),$$

machine-checked as `WeightBound.crit_of_band_geometry` with the per-band identity as `factor_below` and `factor_above`. The criterion then reads entirely in the band ranges of the abelian cover, in the same currency as J_+/J_-.

That bound is not sufficient on its own, and the failure is measured. The only thing discarded is the correlation between bands, since $\sup \prod \leq \prod \sup$ is an equality only when the extrema align, and that costs a median factor 4.73 and up to 28.7. Over 208 residue points the criterion with the true Λ holds every time, worst margin 2.149, while the criterion with the product in place of Λ holds 207 times and fails once, at $J_+/J_- = 1213$ against a product of 2298. The failing point is the one where x sits closest to the edge of the crossing band, minority measure 0.0171, which is not a coincidence: proximity to a band edge makes J_+/J_- large and also tends to bring a second band close to x, which makes the product large. Decoupling the two halves therefore costs most exactly where the margin is thinnest.

Difficulty is conserved and it is worth saying where it went. A weighted mean lies in the convex hull of the range of λ_{k_0}, which is the band containing x, and that band is not inside $\mathrm{spec}(T)$ precisely because x is in it and outside $\mathrm{spec}(T)$. So the containment is not automatic. What has changed is the shape of the remaining work: one scalar in one interval in place of two integrals of comparable size. Two crude routes to it are already closed. Replacing $|Q|$ by its extremes fails because $\sup |Q| / \inf |Q|$ reaches $5.4 \cdot 10^4$ on the same examples, which is far more than the small minority measure buys. And deforming the polytorus radii, which leaves the constant Fourier coefficient unchanged and so preserves $\mu_G(x)$, cannot help either: at radius one the image of P_x lies on a line and spans an arc of exactly π, while any other radius makes A_G non-Hermitian and opens the image into two dimensions, so the arc only grows and the half-plane criterion is never reached (`code/halfplane.py`).

3 The expected non-backtracking polynomial, and how close the bound comes

4 An expected non-backtracking polynomial

Lemma 8 (Chernyak–Chertkov; Watanabe–Fukumizu). *Let G be any graph, A_s its adjacency matrix with edges signed by $s \in \{\pm 1\}^E$, and D the degree matrix. Then, with s uniform,*

$$\mathbb{E}_s \det\bigl(I + u^2(D - I) - uA_s\bigr) = \sum_M (-u^2)^{|M|} \prod_{v \notin V(M)} \bigl(1 + (d_v - 1)u^2\bigr),$$

the sum being over all matchings M of G. If G is (a, b)-biregular bipartite with $P = a - 1$ and $Q = b - 1$, the right side is $\mu_G(\lambda)$ up to normalization at $\lambda = \sqrt{(1 + Pu^2)(1 + Qu^2)}/u$, and then, with $t = u^2$,

$$\lambda^2 - c = \frac{1}{t} + mt. \tag{1}$$

The identity is not new and we claim none of it. Chernyak and Chertkov [14, Eqs. (2), (6)] prove that for an arbitrary matrix H the average of $\det H(\sigma)$ over $\mathbb{Z}_2$ edge gaugings equals a monomer–dimer sum with weights $w_a = H_{aa}$ and $w_{ab} = -H_{ab}H_{ba}$; taking $H = I + u^2(D - I) - uA$ gives the display, since then $w_a = 1 + (d_a - 1)u^2$ and $w_{ab} = -u^2$. Equivalently, the averaging step alone is the Godsil–Gutman expansion [6], valid for any diagonal matrix in place of $I + u^2(D - I)$, and the right-hand side with those degree-dependent weights is the monomer–dimer partition function $\Xi_G(-u^2, \lambda)$, $\lambda_v = 1 + (d_v - 1)u^2$, of Watanabe and Fukumizu [17, Thm. 5.5]; on a $(q+1)$-regular graph they reduce it to $u^{|V|}\mu_G(1/u + qu)$ [17, Eq. (5.3)], attributing the regular case to Nagle, an attribution we take from [17], not having consulted the original, and record the consequent band statement, that the roots then lie on the circle of radius $1/q$. Only (1), the biregular substitution, is not in those sources.

What we do wish to point out is how the left-hand side should be read. By the Ihara–Bass formula for signed graphs, $\mathbb{E}_s \det(I - uW_s)$ is $(1-u^2)^{|E|-|V|}$ times the left side above, so the polynomial in Lemma 8 is an *expected non-backtracking characteristic polynomial*. This reading is absent from the sources just cited, and not by oversight in an obvious direction: [14] obtain the average but explicitly set the Ihara spectral parameter to zero, and [17] display the Bass matrix $I - uA_G + u^2(D_G - I)$ next to their Theorem 5.9, a sum over deletions of vertex-disjoint cycles, and remark in print that its relation to the graph zeta function is unknown. The sign average is exactly what annihilates those cycle terms. The closest antecedent we have found is Watanabe and Chertkov [15, Thm. 4], who do average a non-backtracking determinant over ± 1 edge signs and convert it by Ihara–Bass, but for the permanent, that is for perfect matchings on $K_{N,N}$, with no degree-dependent vertex weights.

The band condition has a clean reading on that polynomial, and it sharpens a published bound. By (1), a root β of ω_G coming from a nonzero root λ of μ_G satisfies $m\beta^2 - \rho\beta + 1 = 0$ with $\rho = \lambda^2 - c$, so the two roots of that quadratic have product $1/m$; they are complex conjugates, hence both of modulus $m^{-1/2}$, exactly when $|\rho| \le 2\sqrt{m}$, and otherwise real with one inside and one outside that circle. Watanabe and Fukumizu prove [17, Cor. 5.8] that every root of ω_G lies in the annulus $1/(d_M - 1) \le |\beta| \le 1/(d_m - 1)$, which for an (a, b)-biregular graph reads $1/(a-1) \le |\beta| \le 1/(b-1)$. Conjecture 1 at $d = 1$ asserts that all roots except those coming from $\lambda = 0$ lie on the single circle $|\beta| = m^{-1/2}$, the geometric mean of the two radii they bound by. The roots at $\lambda = 0$ are $\beta = -1/(a-1)$ and $\beta = -1/(b-1)$, so they sit at the two ends of that annulus, which is why it is attained and cannot be narrowed without excluding them.

Since $1/t + mt = 2\sqrt{m}\cos\varphi$ exactly when $|t| = 1/\sqrt{m}$:

Corollary 9. *For (a, b)-biregular bipartite G, Conjecture 1 at $d = 1$ holds for G if and only if every root u of the expected non-backtracking characteristic polynomial of G, other than the trivial ones, satisfies $|u| = m^{-1/4}$.*

So the conjecture is a Ramanujan condition after all, but for an *expected* polynomial rather than for a spectrum. That is forced. If μ_G were, up to the substitution and prefactor above, the determinant of a single locally described operator on G, its coefficients could be recovered by interpolation from polynomially many determinant evaluations; but those coefficients are the matching numbers m_k, whose computation is $\#P$-hard even for planar graphs of bounded degree [16]. No unaveraged Bass-type identity for the matching polynomial can therefore exist unless FP = #P. This is also why the roots of μ_G are not the spectrum of any operator on G, and hence why the available technology for such statements is the method of interlacing families [23], which works with expected polynomials, rather than a spectral argument.

5 How close the bound comes to being attained

Proposition 19 also explains why the interval is not filled. The mean square of the roots is $\big[(b-2)^2 + a(b-1)\big]$, so the fraction of the band $[-2\sqrt{m}, 2\sqrt{m}]$ occupied in mean square is

$$\frac{(b-2)^2 + a(b-1)}{4(a-1)(b-1)},$$

which is 7/16 for $(a, b) = (3, 3)$, 1/2 for $(2, 2)$, 3/8 for $(4, 3)$, and tends to 1/4 as $a \to \infty$. The bulk of the spectrum sits well inside the band; only the extremes are at issue, which is why moments cannot decide the question.

The extremes behave as Theorem 16 predicts. Writing the least and greatest roots as percentages of the two band edges:

G	girth	least root	greatest root
$K_{3,3}$	4	89.6%	57.2%
$K_{4,4}$	4	94.6%	56.6%
circulant $6+6$	4	96.4%	74.2%
Heawood	6	97.5%	79.9%
Pappus	6	98.6%	83.4%
C_{12}	6	96.6%	96.6%
$3C_4$	4	70.7%	70.7%

The first block has $b = 3$, and the least root is consistently far closer to its edge than the greatest is to its, the roots are displaced downwards by $b-2$, exactly as Theorem 16 requires. The second block has $b = 2$, the displacement vanishes, and the two ends are used equally. That is what the table shows, and it is all it shows.

It does not bear on sharpness, for a reason worth stating because it is easy to miss. Every graph in the table has $a = b$, and there $c = 2(a-1) = 2\sqrt{m}$, so $\rho = y - c \geq -c = -2\sqrt{m}$ holds automatically from $y = x^2 \geq 0$: the lower bound is an identity, not a theorem. This is the triviality that Song, Fan and Miao note for $d = r$ [10]. The percentages above therefore measure only how close the least root of μ_G comes to 0, and 100% would require $\mu_G(0) = 0$, that is a graph with no perfect matching, which by Kőnig's theorem no regular bipartite graph is. The conjecture has content exactly when $a \neq b$, where $c > 2\sqrt{m}$ strictly by the arithmetic–geometric mean inequality.

Where it has content the bound holds with room to spare, and is not approached. For $(4, 2)$-biregular graphs the conjecture asserts $y \geq c - 2\sqrt{m} = 4 - 2\sqrt{3} = 0.5359$; a girth-4 family plateaus at $y_{\min} = 1.1716$ and a girth-8 family at 0.7392. In the trivial regime the same plateauing is visible, and there the limit is an algebraic number one can name. For a cubic bipartite circulant on $n+n$ vertices with connection set S, the matching counts satisfy $f_n(y) = \operatorname{tr} A(y)^n$ for a transfer matrix $A(y)$ with entries linear in y, so by the theorem of Beraha, Kahane and Weiss the roots accumulate where the two dominant eigenvalues of $A(y)$ have equal modulus. That happens by a genuine collision of two real eigenvalues, below y^* the pair is real and distinct, above it a complex-conjugate pair, so the locus is cut out by $\operatorname{disc}_\lambda \chi_{A(y)} = 0$ and y^* is the left edge of the accumulation band. For $S = \{0, 1, 2\}$, girth 4,

$$\operatorname{disc}_\lambda \chi = 4y^2\big(y^4 - 12y^3 + 46y^2 - 84y + 5\big),$$

and y^* is the least positive root of that quartic, which is irreducible over $\mathbb{Q}$; then $y^* = 0.0615663466041589\ldots$ and the limit is $100 - 25y^* = 98.4608413349\ldots\%$ of the lower edge. For $S = \{0, 1, 3\}$, girth 6, the discriminant is $y^3F(y)$ with F irreducible of degree 17, giving $y^* = 0.0059255959422562\ldots$ and $99.8518601014\ldots\%$. Both stop strictly short of 100%. The finite-n values approach these from above with a $1/n^2$ band-edge law, and $n^2\big(y_{\min}(n) - y^*\big)$ is constant to five digits at $n = 200, 300, 500$, which identifies the extrapolated plateau with the algebraic number.

One half of the limit is certified exactly, without appeal to Beraha–Kahane–Weiss. On the whole of $[0, y^*)$ the largest eigenvalue modulus of $A(y)$ is attained by exactly one eigenvalue, which is real, simple and negative: the discriminant has no zero there, so real branches stay real; no two eigenvalues are negatives of one another, by a resultant of the even and odd parts of χ which has no zero on $(0, y^*]$; and for the octic a separating radius $5/4$, which no eigenvalue ever attains, together with one exact Schur count, pins the number of eigenvalues outside it at two. Hence $|f_n| \geq |\lambda_1|^n(1 - (N-1)(1+\delta)^{-n})$ on any compact subinterval, giving $\liminf_n y_{\min}(n) \geq y^*$. The reverse inequality does use Beraha–Kahane–Weiss: just above y^* the two branches form a conjugate pair of equal and maximal modulus, and one needs that theorem to conclude that the zeros accumulate there. This is what one should expect: a family with a fixed local structure has a fixed Benjamini–Schramm limit, which is not a tree, so its matching measure retains a gap at 0. Non-attainment of this kind was already visible to Heilmann and Lieb, who tabulate the largest zero against their own bound for six lattices, 11.24 against 12, 17.86 against 20, 41.32 against 44, and remark that equality "cannot be expected to be true in general" [13, Table 1 and Remark 4.3]. Girth does help monotonically, at 30 vertices, girth 4, 6 and 8 give 98.166%, 99.311% and 99.410%, but no family of bounded girth can approach the edge, and every cubic bipartite circulant has girth at most 6, since the closed hexagon $0, s_1, s_1 - s_2, s_1 - s_2 + s_3, s_3 - s_2, s_3$ is always present. We therefore make no claim that $2\sqrt{m}$ is attained or approached. The roots of these circulants do satisfy (5), with maxima $4.06, 4.48, 6.26$ against bounds $4.90, 5.66, 6.93$.

6 The coordinate case at rank b, and the sharpness of Xus bound

6.1 The coordinate case at rank b, and the sharpness of Xu's constant

Theorem 23 identifies the recentred mixed characteristic polynomial with the matching polynomial in the commuting case at $b = 2$. That identification survives to every rank, and the object it produces is again an ordinary matching polynomial, on a different graph. The consequence is that Xu's constant is not merely an upper bound at $b \geq 3$: it is exactly the truth, and the class attains it.

Let H be an a-regular b-uniform hypergraph on n vertices with q hyperedges, and let P_e be the coordinate projection onto $\operatorname{span}\{e_v : v \in e\}$. These have rank b and satisfy $\sum_e P_e = aI$ exactly, so they are an admissible tight family. Write $I(H)$ for the incidence bipartite graph, with one side the n vertices and the other the q hyperedges; it is (a, b)-biregular.

Proposition 10. *With μ the mixed characteristic polynomial of $\{P_e\}$,*

$$\mu(x) = \sum_{k \geq 0} (-1)^k\, m_k\big(I(H)\big)\, x^{n-k}, \qquad \textit{equivalently} \qquad \mu(t^2) = t^{n-q}\, \mu_{I(H)}(t),$$

where m_k is the number of k-matchings. In particular the positive roots of μ are exactly the squares of the positive roots of $\mu_{I(H)}$.

Proof. The matrix $xI + \sum_e z_e P_e$ is diagonal, with v-th entry $x + \sum_{e \ni v} z_e$, so $\det = \prod_v \big(x + \sum_{e \ni v} z_e\big)$. Each factor is affine in every z_e, so for $S \subseteq [q]$ the derivative $\partial_S \det$ evaluated at $z = 0$ is obtained by choosing, for each $e \in S$, one factor to differentiate, that is one vertex $v \in e$; two hyperedges choosing the same vertex differentiate one affine factor twice and contribute 0. Hence $\partial_S \det$ at $z = 0$ counts the injective maps $S \to V$ with $e \mapsto v \in e$, each contributing $x^{n-|S|}$. Applying $\prod_e (1 - \partial_{z_e})$ and grouping by $|S| = k$, the coefficient is the number of systems of distinct representatives of k hyperedges, which is the number of k-matchings of $I(H)$ saturating k hyperedge-nodes. The second display follows from $\mu_{I(H)}(t) = \sum_k (-1)^k m_k t^{n+q-2k}$ on substituting $x = t^2$. $\square$

Corollary 11. *For every coordinate family as above, at every $b \geq 2$,*

$$\operatorname{maxroot} \mu \ \leq \ \big(\sqrt{a-1} + \sqrt{b-1}\big)^2.$$

Proof. $I(H)$ is (a, b)-biregular, so its universal cover is the (a, b)-biregular tree, of spectral radius $\rho = \sqrt{a-1} + \sqrt{b-1}$. Godsil's bound [26] puts every root of $\mu_{I(H)}$ in $[-\rho, \rho]$, and Proposition 10 squares them. $\square$

The coordinate hypothesis is basis-dependent as stated and is not meant to be. It says exactly that the family commutes.

Corollary 12. *Let $a, b \geq 2$ and let $P_1, \ldots, P_q$ be pairwise commuting rank-b orthogonal projections on $\mathbb{R}^n$ with $\sum_k P_k = aI$. Then* $\operatorname{maxroot} \mu \leq (\sqrt{a-1} + \sqrt{b-1})^2$*, and the constant is attained in the limit over this class.*

The hypothesis $a, b \geq 2$ is needed and not cosmetic. At $a = b = 1$, $n = q = 1$, $P_1 = [1]$, one has $\mu(x) = x - 1$ and $\operatorname{maxroot} \mu = 1$ against a right-hand side of 0. The bound is a statement about the (a, b)-biregular tree, and that tree is a single edge or a ray unless both degrees are at least two.

Proof. Commuting orthogonal projections are simultaneously diagonalisable, and in a common eigenbasis each P_k is a 0/1 diagonal matrix with exactly b ones, that is a hyperedge e_k on n vertices; $\sum_k P_k = aI$ says every vertex lies in exactly a of them. So the family is the coordinate family of an a-regular b-uniform hypergraph, and Corollary 11 and Proposition 13 apply. Both statements are invariant under a common orthogonal change of basis, so nothing is lost in choosing one. $\square$

So the inequality Xu conjectures is a theorem on the commuting locus, at every rank, and not only at $b = 2$ where it is Heilmann–Lieb. The question that matters for the conjecture is whether the constant is the right one there, and it is.

Proposition 13. *Fix $a, b \geq 2$. Over coordinate families with parameters (a, b),*

$$\sup \ \frac{\operatorname{maxroot} \mu}{\big(\sqrt{a-1} + \sqrt{b-1}\big)^2} \ = \ 1.$$

Proof. At most 1 by Corollary 11. For the other direction, let B_r be the ball of radius r in the (a, b)-biregular tree. If girth $I(H) > 2r$ then the r-ball of $I(H)$ about any a-side vertex is acyclic and, by biregularity, isomorphic to B_r, so $B_r \subseteq I(H)$ as a subgraph. The greatest root of a matching polynomial is monotone under subgraphs [13, 26], and B_r is a tree, where the matching polynomial is the characteristic polynomial; hence maxroot $\mu_{I(H)} \geq \lambda_{\max}(B_r)$, and squaring, maxroot $\mu \geq \lambda_{\max}(B_r)^2$. The balls exhaust the tree and $\lambda_{\max}(B_r) \uparrow \rho$. Finally (a, b)-biregular bipartite graphs of arbitrarily large girth exist: random N-lifts of any one of them preserve biregularity and have girth tending to infinity in probability. Taking $r \to \infty$ gives the claim. □

The ladder is measured rather than assumed, and it also supplies an explicit floor for any graph whose girth is known. Writing the ratio in the μ variable, $\lambda_{\max}(B_r)^2/\rho^2$ is

(a, b)	$r = 2$	4	6	8	10	12
$(3, 2)$	0.6863	0.8579	0.9167	0.9443	0.9596	0.9692
$(4, 3)$	0.6061	0.8082	0.8854	0.9234	0.9450	0.9585
$(5, 4)$	0.5744	0.7898	0.8747	0.9168	0.9396	
$(4, 4)$	0.5833	0.7951	0.8777	0.9186	0.9418	0.9505
$(6, 3)$	0.6004	0.8031	0.8826	0.9219	0.9442	0.9489

increasing in r at every pair, $b \geq 3$ included; the missing entry is where the ball passed the 400,000 vertex cap, and a truncated ball is a proper subtree of the true one, hence still a subgraph of any graph of girth $> 2r$, so the floor it gives remains valid. On named incidence graphs, with girth recomputed by breadth-first search rather than quoted, the realized ratios are 0.8995 for Heawood, 0.9171 for Pappus and 0.9453 for Tutte–Coxeter at $(3, 3)$; 0.9098 for AG$(2, 3)$ at $(4, 3)$; and 0.9097 for PG$(2, 3)$ at $(4, 4)$. Every one exceeds its ball floor, which is the containment argument of Proposition 13 checked against the object it is about (`code/xu_sharp.py`).

Two things follow, and the second is the reason for the subsection.

First, the plateaus found by the adversarial search of `code/adversarial.py` and `code/rl_push.py` — 0.739 at $(4, 3)$ and 0.726 at $(5, 4)$, against 0.939 at $(3, 2)$, none of them a violation — are an artefact of that search, not a property of the class. It sampled generic subspaces, and generic subspaces behave like short girth. The coordinate construction reaches 0.9098 at $(4, 3)$ on twenty-one vertices and is bounded below by 0.9585 at girth 26, well past where the search stalled. There is no evidence of a smaller true constant at $b \geq 3$, and the reading that suggested one is closed.

Second, the constant is now fixed. Any proof of Xu's conjecture in general must produce exactly $(\sqrt{a-1} + \sqrt{b-1})^2$; any attempted improvement of it is false, by Proposition 13, before it is attempted. Equally, the whole content of the conjecture lies off the commuting locus: the bridge of Proposition 10 fails for non-coordinate rank-b projections, which is checked as a control rather than asserted, and it is that failure which leaves the general statement open.

Whether that leaves the general statement hard or merely unreduced turns on one further question, which is measurable: is the commuting locus *extremal*? Deform a commuting family by rotating each block independently, $U_e = \exp(tX_e)$ with X_e skew, and return to the tight projection class by alternating projections onto {rank-b projections} and $\{\sum_k A_k = aI\}$. Along such a path the ratio falls at every hypergraph tried; an ascent started at the commuting point accepted none of 400 steps at any of four starting points; and, which is the part that bears on a global statement, no random start in the tight projection class beat its commuting counterpart at the same (n, q, a, b), over fifteen starts each, the best random family falling short by between 0.0038 and 0.0107 (`code/offcoord.py`). On that evidence Conjecture 1.4 would follow from Corollary 12 together with the statement that the commuting locus is extremal, which replaces a conjecture about all tight rank-b families by a monotonicity statement over a theorem.

Proposition 19 says which directions are excluded; the same computation says which are admitted, and the answer is an explicit cone.

Call i and j *twins* when they lie in exactly the same hyperedges, and the family *twin-free* when it has no twin pair. Twin-freeness is checkable in $O(n^2 q)$ and holds for every family used below; the Fano family, C_4 and the triples of K_4 are all twin-free.

Proposition 14. *Let D lie in the kernel of the linearised constraints. A second-order correction X exists if and only if*

$$Q_j(D) := \sum_k \sigma_k(j)\,(D_k^2)_{jj} = 0 \qquad \text{for every vertex } j,$$

with $\sigma_k(j)$ as in Proposition 19, together with

$$\sum_k \sigma_k(i,j)\,(D_k^2)_{ij} = 0 \qquad \text{for every twin pair } \{i,j\}, \qquad \sigma_k(i,j) = 1 - (P_k)_{ii} - (P_k)_{jj}.$$

For a twin-free family the second family of conditions is empty, and the directions admitting a second-order correction are then the kernel of the linearisation cut by the n quadrics Q_j.

Proof. The order-two equation $X_k - P_k X_k - X_k P_k = D_k^2$ has (i,j) entry $(X_k)_{ij}\big(1 - (P_k)_{ii} - (P_k)_{jj}\big) = (D_k^2)_{ij}$, so the coefficient is -1 when both i, j lie in e_k, $+1$ when neither does, and 0 when e_k splits the pair. Where the coefficient is nonzero $(X_k)_{ij}$ is forced; where it vanishes the entry is free and the equation demands $(D_k^2)_{ij} = 0$, which holds automatically since D_k is supported on split pairs and D_k^2 therefore is not. Imposing $\sum_k X_k = 0$ entry by entry, a pair is solvable exactly when some e_k splits it, leaving a free entry to absorb the forced sum, or else the forced values already cancel. No e_k splits a diagonal pair, which gives the conditions $Q_j(D) = 0$; and no e_k splits an off-diagonal pair exactly when its two vertices are twins, which gives the second family. □

The entrywise criterion is machine-checked as `ConeFix.solvable_iff`, together with `coef_eq_zero_iff` for the combinatorial reading of the coefficient and `no_free_entry_iff_twin` for the identification of the obstructed pairs as the twin pairs. The twin conditions are not vacuous. On the 3-uniform 2-regular family with vertices $\{0, \ldots, 5\}$ and hyperedges $\{0,1,2\}, \{0,1,3\}, \{2,4,5\}, \{3,4,5\}$, where 0 and 1 are twins, the direction $D_1 = E_{04} + E_{40} + E_{14} + E_{41}$, $D_2 = 0$, $D_3 = -(E_{04} + E_{40})$, $D_4 = -(E_{14} + E_{41})$ lies in the kernel and satisfies $Q_j(D) = 0$ at all six vertices, while the forced values at the twin entry $(0, 1)$ sum to -1: no second-order correction exists. Every step of that witness is machine-checked in `RamaLean/ConeWitness.lean` (`twins`, `cross_supported`, `sum_D_eq_zero`, `Q_eq_zero`, `twin_obstruction`), so the necessity of the twin conditions rests on no arithmetic done outside the kernel.

Proof of the twin-free case, restated. Necessity is the computation in Proposition 19: the diagonal of each X_k is forced and summing it over k gives $Q_j(D)$, which $\sum_k X_k = 0$ requires to vanish. For sufficiency, every off-diagonal pair is split by some e_k and the free entry there absorbs the forced sum, so only the diagonal conditions remain. □

The second-order behaviour of the top root over that cone can now be examined, and it is better than the cone alone would suggest. At a critical point the second derivative of $\lambda_{\max}$ along a curve in V depends only on the curve's velocity, so it defines a quadratic form Λ on the admissible directions; and since the 2-jet $P_k + \varepsilon D_k + \varepsilon^2 X_k$ agrees with a curve on V to $O(\varepsilon^3)$, Λ may be read off that jet without constructing the curve. Computed this way on the Fano family, by fitting the coefficients of μ as polynomials in ε rather than differencing them, Λ is negative semidefinite on the whole linearised kernel, with 42 negative eigenvalues, 21 zero and none positive, and its null space is exactly the tangent space to the orthogonal conjugation orbit (`code/hessian.py`). The value on a cross direction agrees to eight digits with the exact rational $-C/4$ of Proposition 18's computation.

Two things follow. First, no argument restricted to the cone is needed for the second-order test: $\Lambda \preceq 0$ already on the kernel, so it is negative on the cone a fortiori, strictly away from the conjugation directions along which μ does not change at all. Second, the commuting family is a strict local maximum of $\lambda_{\max}$ to second order, transverse to its own orbit.

The signature is certified rather than observed. Each entry of the form is an exact integer polynomial in y: the jet has integer entries when D does, which the basis arranges, so the coefficients of μ are integer polynomials in ε and the ε^2 part is extracted by truncated series arithmetic over $\mathbb{Z}$. The vanishing on the orbit needs no arithmetic at all, since μ is invariant under simultaneous orthogonal conjugation and $\lambda_{\max}$ is therefore constant along $\exp(\varepsilon\Omega)P_k\exp(-\varepsilon\Omega)$; the computation returns exactly 0 there. On a complement of the orbit the restricted form has eigenvalues in $[-0.1269, -0.0218]$ and its negative admits a Cholesky factorisation with smallest pivot 3.7×10^{-2}, against an evaluation error of order 10^{-55} at the working precision (`code/curvform.py`). The margin exceeds the error by some 10^{53}, so the definiteness is a certificate and not a reading.

The extra null directions that appear at $b = 2$ are not a failure of maximality. Beyond the conjugation orbit the form has 2 further null directions for C_4, 6 for K_4 and 9 for C_6, all exact, and none at $(2,3)$, $(3,3)$ or the family of triples in K_4; so the trigger is $b = 2$ and not $a = b = 2$, K_4 and $K_{3,3}$ being $(3,2)$. Along those directions $\lambda_{\max}$ still decreases, at fourth order rather than second, with the same sign for both signs of the parameter: the fourth-order coefficients measured are -0.0038 and -0.0005 for C_4 and between -0.60 and -0.14 for K_4. At $b = 2$ the second-order test is blind and the fourth order does the work.

It is not a statement about all commuting families, and the qualification is not idle. The same computation on four further families gives no positive eigenvalue anywhere, so $\Lambda \preceq 0$ throughout; but the strictness transverse to the orbit fails at $(a,b) = (2,2)$, where for C_6 the null space has dimension 24 against an orbit of rank 15, nine directions carrying no curvature without being conjugations. It holds at $(2,3)$ with smallest pivot 8.7×10^{-2} and at $(3,3)$, and is vacuous for the family of triples in K_4, whose whole kernel is the orbit. The $(2,2)$ case is the one whose universal cover is a line, with no spectral gaps at all, so its degeneracy is not surprising; but on the evidence the correct statement is $\Lambda \preceq 0$ in general, with strictness established only at the families where it was certified. The excess is triggered by $b = 2$ and not by $a = b = 2$: K_4 and $K_{3,3}$ are $(3,2)$ and carry excess 6 and 9, as the table above records.

One order beyond the obstruction, nothing further is imposed, and for a reason of supports. The order-three equation reads $Y_k - P_k Y_k - Y_k P_k = D_k X_k + X_k D_k$. Its off-diagonal blocks give the condition $(D_k X_k + X_k D_k)_{12} = 0$, which holds identically: with $X_{11} = -(D^2)_{11}$ and $X_{22} = (D^2)_{22}$ from order two, that block is $D_{12}D_{21}D_{12} - D_{12}D_{21}D_{12} = 0$. Its diagonal entries vanish too. The diagonal is where the content is. No e_k splits a diagonal pair, so $(Y_k)_{jj}$ is forced and $\sum_k Y_k = 0$ requires

$$R_j := \sum_k \sigma_k(j)\,(D_k X_k + X_k D_k)_{jj} = 0 \qquad \text{for every } j.$$

R is not identically zero: at the Fano family a second-order correction satisfying $\sum_k X_k = 0$ gives $\max_j |R_j| = 19.7$. What is identically zero is its *sum*, and that is a two-line theorem holding per block, with no hypothesis on X at all.

Lemma 15 (order-three trace law)**.** *If $P_k D_k + D_k P_k = D_k$ then $\operatorname{tr}(D_k X_k + X_k D_k) = 2\operatorname{tr}\big(P_k(D_k X_k + X_k D_k)\big)$ for every X_k, that is $\sum_j \sigma_k(j)(D_k X_k + X_k D_k)_{jj} = 0$. Summing over k, $\sum_j R_j = 0$.*

Proof. $\operatorname{tr}(P_k(D_k X_k + X_k D_k)) = \operatorname{tr}(P_k D_k X_k) + \operatorname{tr}(P_k X_k D_k) = \operatorname{tr}(P_k D_k X_k) + \operatorname{tr}(D_k P_k X_k) = \operatorname{tr}((P_k D_k + D_k P_k) X_k) = \operatorname{tr}(D_k X_k)$ by cyclicity and the order-one equation, while $\operatorname{tr}(D_k X_k + X_k D_k) = 2\operatorname{tr}(D_k X_k)$. □

Machine-checked as `OrderThreeLaw.trace_order_three`. It needs neither idempotency, nor tightness, nor any support hypothesis, and it replaces a complementary-supports argument given in an earlier version of this note whose hypothesis on X is satisfied by no second-order correction: at the Fano family the canonical X with vanishing split-pair entries fails $\sum_k X_k = 0$ at all 21 of 21 off-diagonal pairs, those entries being the only free ones and hence the only ones that can absorb the forced same-side sums.

The second-order correction is not unique, its split-pair entries being free, and R depends on the choice. So order three is unobstructed exactly when some admissible X makes $R = 0$. Lemma 15 puts R in the hyperplane $\sum_j R_j = 0$ for every choice, and the image of the free entries is the same object that governs the second order.

Proposition 16. *Write u for the free entries of X, which satisfy $\sum_{k \in K(i,j)} u_{k,i,j} = 0$ pair by pair. The map $u \mapsto R$ is linear with no constant term and has rank $n - c(G_D)$, with G_D the graph of Proposition 14's companion: $i \sim j$ when $(\sigma_k(j) D_k(i,j))_{k \in K(i,j)}$ is not constant in k. Consequently, if G_D is connected then order three is unobstructed.*

Proof. R is linear in u because D_k is supported exactly on the split pairs, which are exactly the free entries. Using $\sigma_k(i) = -\sigma_k(j)$ on a split pair,

$$\sum_j w_j R_j = 2 \sum_{i<j} (w_j - w_i) \sum_{k \in K(i,j)} \sigma_k(j) D_k(i,j) u_{k,i,j},$$

so w annihilates the image exactly when, pair by pair, either $w_i = w_j$ or the linear functional $u \mapsto \sum_k \sigma_k(j) D_k(i,j) u_k$ vanishes on $\{\sum u = 0\}$, which happens exactly when its coefficients are constant in k. That is the definition of G_D, and the annihilator is therefore the functions constant on its components, of dimension $c(G_D)$. If G_D is connected the rank is $n-1$, which is the dimension of the hyperplane $\sum_j R_j = 0$ that Lemma 15 confines R to, so the image is that hyperplane and $R = 0$ is attained. □

Every step is machine-checked: `OrderThreeLaw.trace_order_three` for the lemma, `SpanRank.linear_vanishes_iff_const` for the annihilator condition, `CokernelRank.const_of_adj_eq` for the passage from edges to components, and `OrderFourSolvable.mem_of_sum_zero` for the conclusion. The rank formula is verified against the computed rank at five families and twenty cone directions, and order three is solvable at every one of them, including C_4, whose G_D has two components so that the proposition does not apply and solvability there is measured (`code/orderthree.py`).

Order four is then satisfied as well, and the freedom in Y is what absorbs it. The condition is $\sum_k \sigma_k(j)\big[2(D_kY_k)_{jj} + (X_k^2)_{jj}\big] = 0$ at each vertex, linear in Y, so it holds exactly when the X^2 vector lies in the image of $L_D : Y \mapsto \big(2\sum_k \sigma_k(j)(D_kY_k)_{jj}\big)_j$. That image always lies in the trace-zero hyperplane, and for the same kind of reason: writing the corners of D_k and Y_k as M and N, the two halves of $\operatorname{tr}\big((I-2P_k)D_kY_k\big)$ are $\operatorname{tr}(MN^\mathsf{T})$ and $\operatorname{tr}(M^\mathsf{T}N)$, which are equal, so the sum over j of the image vanishes identically (`OrderFour.image_in_trace_zero`). That it fills that hyperplane has a criterion. A vector w annihilates the image exactly when $(w_i - w_j)\big(\sigma_k(i)D_k(i,j) - \sigma_l(i)D_l(i,j)\big) = 0$ for every pair i,j and every pair of blocks k,l separating them, so $w_i = w_j$ as soon as two separating blocks disagree at (i,j). Joining i to j whenever they do, w is constant on each component and

$$\operatorname{rank} L_D = n - c(G_D),$$

with c the number of components. The cokernel is therefore spanned by $(1,\dots,1)$, and order four is solvable, exactly when G_D is connected. Checked against the computed rank on 32 directions across two families, agreeing in every case, the cross basis directions being precisely those where G_D is empty and the rank collapses to 1; there the obstruction vanishes outright and order four is solvable regardless. The scalar step and the passage from edges to components are machine-checked as `CokernelRank.sep_forces_eq` and `CokernelRank.const_of_adj_eq`. And the X^2 vector lies in it for a reason. Summing over the vertices,

$$\sum_j \sum_k \sigma_k(j)(X_k^2)_{jj} = \sum_k \Big[\operatorname{tr}\big((D_k^2)_{22}^2\big) - \operatorname{tr}\big((D_k^2)_{11}^2\big)\Big],$$

the contributions of the free block X_{12} cancelling through $\operatorname{tr}(X_{21}X_{12}) = \operatorname{tr}(X_{12}X_{21})$. Since D_k is off-diagonal for the splitting, writing its corner as M gives $(D_k^2)_{11} = MM^\mathsf{T}$ and $(D_k^2)_{22} = M^\mathsf{T}M$, whose squares have equal trace by cyclicity, so every term vanishes. The identity is machine-checked as `OrderFour.order_four_sum_zero`, and the residuals of the order-four solve are 10^{-18} on the directions tested (`code/tangentcone.py`). Those two facts, that the obstruction sums to zero and that the cokernel is the constants, do not by themselves put the obstruction in the image, and the step that does is `OrderFourSolvable.order_four_solvable`: in finite dimension a subspace whose orthogonal complement is the line of constants is the hyperplane of vectors summing to zero. It had been left unstated between two files that each proved half of it.

Orders two, three and four are therefore all satisfiable on the cone. The passage from every finite order to an actual curve was the remaining gap, and it can be closed rather than climbed, because the variety is in the scope of a classical criterion.

Beyond the second order, the extremality itself has been searched for directly. At four commuting families the commuting point is the largest value found over between 32 and 40 random tight projection families at the same (p,q,a,b), by margins of 0.0004 for C_4, 0.016 for C_6, 0.046 for $K_{3,3}$ and 0.048 for the cube, every value staying below the band edge (`code/extremal.py`). That is the statement of §D seen from the other side: within the projection class the commuting families are the largest, while across the rank-n interpolation the projections are the smallest. Neither observation is a proof, and the conjunction of the two is what Conjecture 1.4 would need.

Proposition 17 (the curve, by 2-regularity)**.** *Let D lie in the kernel of the linearisation with $Q(D) = 0$, and suppose the linear map $E \mapsto \mathrm{d}Q(D)[E]$ carries that kernel onto the span of $Q_1, \ldots, Q_n$. Then there is a curve $t \mapsto A(t)$ in the tight projection variety with $A(0) = A_0$ and $A(t) = A_0 + tD + O(t^2)$. In particular D lies in the tangent cone.*

Proof. Work upstairs, on frames $U_k \in \mathbb{R}^{n\times b}$ with $U_k^{\mathsf{T}} U_k = I_b$ and $A_k = U_k U_k^{\mathsf{T}}$, so that idempotency is free and the only equation is $\Psi(U) = \sum_k U_k U_k^{\mathsf{T}} - aI = 0$. The Stiefel constraint makes the e_k-block of a first-order frame variation antisymmetric, and a diagonal entry of $\mathrm{d}\Psi$ is a diagonal entry of that block, so $\mathrm{d}\Psi$ has vanishing diagonal; and it is onto the zero-diagonal symmetric matrices, of rank 21 against 28 at the Fano family. Split Ψ accordingly. Its off-diagonal part is a submersion, so $W = \{\Psi_{\text{off}} = 0\}$ is a smooth manifold near the point, and on W the remaining map $F = \Psi_{\text{diag}}$ satisfies $F = 0$ and $\mathrm{d}F = 0$ there, with Hessian Q.

Write $F = Q + C + \cdots$ with $Q(x) = B(x, x)$ quadratic and C cubic. Since $Q(D) = 0$, the polynomial $t \mapsto F(tD + t^2 w)$ vanishes to order three, so $F(tD + t^2 w) = t^3 G(t, w)$ with G polynomial and $G(0, w) = 2B(D, w) + C(D, D, D)$, affine in w with surjective linear part by hypothesis. Pick w_0 with $G(0, w_0) = 0$ and apply the implicit function theorem in w at $(0, w_0)$; the resulting $x(t) = tD + t^2 w(t)$ satisfies $F(x(t)) = 0$ and $x(t)/t \to D$. □

This is Avakov's 2-regularity condition [28, 29] in the fully degenerate case, and no novelty is claimed for it. What is new here is only that the tight projection variety is in its scope, which is what the frame reduction supplies.

The span in the hypothesis is not all of $\mathbb{R}^n$, and the reason is an identity that holds for each block separately,

$$\sum_j \sigma_k(j)(D_k^2)_{jj} = 0,$$

since $(D_k^2)_{jj} = \sum_l D_k(j, l)^2$ by symmetry, so that exchanging the two indices in the double sum pairs every surviving term with one of opposite sign, D_k being supported on pairs with exactly one endpoint in e_k. So Q lands in the trace-zero diagonals, and the span can be smaller still: it is $n - 1$ at C_6, $K_{3,3}$ and the Fano family, and 2 at C_4, which carries the extra pair $Q_0 + Q_2 = Q_1 + Q_3 = 0$, one for each side of its bipartition. The identity is machine-checked as `ArcTangent.sum_sgn_diag_sq_eq_zero`, and the passage from a curve to membership of the tangent cone as `ArcTangent.mem_tangentCone_of_arc`; the implicit function theorem step is not formalised.

The hypothesis is a rank condition on D alone, and it holds where it has been checked: at every cone direction tested at C_4, $K_{3,3}$ and the Fano family, four directions each, the rank of $\mathrm{d}Q(D)$ on the kernel equals the span. The curve has also been exhibited rather than inferred, by continuation upstairs: the residual stays at 10^{-14} or below and $\|(A(t) - A_0)/t - D\|$ falls with exponent 0.96 to 1.00 over $t \in [0.01, 0.05]$, while a direction with $Q(D) \neq 0$ has the same quantity flat at exponent 0 (`code/arc.py`).

Both quantities in that hypothesis are combinatorial, and saying so answers the question of when it holds. Write $K(i,j)$ for the blocks separating i from j. A kernel direction has D_k supported on separated pairs and tightness reads $\sum_{k\in K(i,j)} D_k(i,j) = 0$ pair by pair, with distinct pairs independent; and since $(D_k^2)_{jj} = \sum_l D_k(j,l)^2$ and $\sigma_k(i) = -\sigma_k(j)$ for separating k,

$$\sum_j w_j Q_j(D) = \sum_{i<j}(w_j - w_i) \sum_{k\in K(i,j)} \sigma_k(j) D_k(i,j)^2, \qquad \sum_j w_j \, \mathrm{d}Q_j(D)[E] = 2\sum_{i<j}(w_j - w_i) \sum_{k\in K(i,j)} \sigma_k(j) D_k(i,j) E_k($$

So in each case w annihilates exactly when, pair by pair, either $w_i = w_j$ or a form vanishes identically on the hyperplane $\{\sum t = 0\}$. A linear functional vanishes there exactly when its coefficients are constant; a diagonal quadratic form $\sum \varepsilon_k t_k^2$ vanishes there exactly when $\varepsilon_a + \varepsilon_b = 0$ for every pair, which for signs ± 1 means $|K(i,j)| \le 1$, or $|K(i,j)| = 2$ with the two separating blocks on opposite sides. Writing G_Q for the graph on the vertices joining i to j when the quadratic form does *not* vanish, and G_D for the graph joining them when $(\sigma_k(j) D_k(i,j))_{k\in K(i,j)}$ is not constant,

$$\operatorname{span}\{Q_1, \dots, Q_n\} = n - c(G_Q), \qquad \operatorname{rank} \mathrm{d}Q(D) = n - c(G_D),$$

with c the component count. G_Q depends on the hypergraph alone, G_D on the direction; G_D is always a subgraph of G_Q, so the rank never exceeds the span, and 2-regularity at D says exactly that the two graphs have the same components. Both formulas agree with the computed values at seven families (`code/spanrank.py`), and the scalar steps are machine-checked in `RamaLean/SpanRank.lean`.

On a tight family the criterion collapses further, and the collapse says exactly when the span drops. Writing λ_{ij} for the number of blocks containing both i and j,

$$|K(i,j)| = \deg(i) + \deg(j) - 2\lambda_{ij} = 2(a - \lambda_{ij}),$$

every degree being a. So $|K(i,j)|$ is even and the two sides of a separated pair carry $a - \lambda_{ij}$ blocks each; the case $|K| = 2$ with both blocks on the same side cannot arise, and $|K| = 2$ always means the cross configuration. The whole criterion becomes a codegree condition,

$$G_Q: \quad i \sim j \iff \lambda_{ij} \le a - 2,$$

so G_Q is the complement of the graph joining pairs of codegree at least $a-1$, and

$$\text{the span drops below } n-1 \iff \text{the codegree-at-least-}(a-1) \text{ graph is a complete join,}$$

a graph being a join exactly when its complement is disconnected. Over a sweep of tight families this happens four times (`code/spanrank.py`): at C_3, whose codegree graph is K_3; at C_4, whose codegree graph is $C_4 = K_{2,2}$, giving the complement $2K_2$ and $c(G_Q) = 2$, which is the exception recorded above; at the four triples of K_4, where every pair lies in two of them, so the codegree graph is K_4, the quadrics vanish identically and the tangent cone is the whole kernel; and at the 2-regular 3-uniform family on six vertices, whose codegree graph is the cocktail party graph $K_{2,2,2}$. For $a = b = 2$ the codegree graph is the cycle itself, and among cycles only C_3 and C_4 are joins, which is why every longer cycle has span exactly $n-1$.

It also settles, in the negative, whether every cone direction is 2-regular. A cross basis direction is supported on a single pair with two separating blocks of opposite sign; the vanishing criterion above is exactly why it satisfies $Q = 0$, so it lies on the cone, and the constancy criterion makes $\operatorname{rank} \mathrm{d}Q(D)$ equal to 0 when $|K(i,j)| = 2$ and to 1 otherwise, against a span of at least 2. Checked at seven families, where the number of such directions runs from 12 at C_4 to 324 at $\mathrm{AG}(2,3)$. Those directions are not lost: Proposition 18's rotation is an explicit curve with exactly that velocity. What is lost is a single uniform route, and the two routes together cover every direction tested.

The earlier evidence, which measured distances rather than curves, is consistent and is recorded because it covers directions the rank condition was not tested on. Over the 42 cross basis directions of the Fano family and over directions obtained by projecting random kernel elements onto $\{Q = 0\}$, the nearest point of the variety to $A_0 + \varepsilon D$ lies at distance $O(\varepsilon^2)$, and at $O(\varepsilon)$ for every direction with $Q(D) \neq 0$ (`code/tangentcone.py`); the general relation is $\operatorname{dist}(A_0 + \varepsilon D, V) = \varepsilon \operatorname{dist}(D, T) + O(\varepsilon^2)$, which is what the two exponents measure. The consequence worth recording is that second-order optimisation over the commuting locus is a constrained problem with n quadratic constraints rather than an unavailable one.

The deformation is also far more rigid than the tangent cone alone suggests.

Proposition 18. *Along the rotation of Proposition 14's cross configuration, the coefficients of $y^n, y^{n-1}, y^{n-2}, y^{n-3}$ in μ are independent of θ, so the θ-dependent part of μ has degree at most $n-4$ in y.*

Proof. Only subsets containing exactly one of e, f carry θ, and they pair as $S \cup \{e\}$ with $S \cup \{f\}$. Write $N_e = P_{e \setminus v} + M_S$ and $N_f = P_{f \setminus w} + M_S$, both diagonal since every block other than e, f is a coordinate projection, and put $u = \cos\theta\, \varepsilon_v + \sin\theta\, \varepsilon_w$, $u^\perp = -\sin\theta\, \varepsilon_v + \cos\theta\, \varepsilon_w$, so that $A_e + M_S = N_e + P_u$ and $A_f + M_S = N_f + P_{u^\perp}$. From $\det(xI + N + P_u) = \det(xI + N) + u^\mathsf{T} \operatorname{adj}(xI + N) u$ and the diagonality of N, whose adjugate has (i,i) entry $\prod_{l \neq i}(x + d_l)$,

$$\det(xI + A_e + M_S) + \det(xI + A_f + M_S) = \big[\theta\text{-free}\big] + \sin^2\theta\, \Delta(x),$$

$$\Delta(x) = (d_v - d_w)\Big[\prod_{l \neq v,w} (x + d_l) - \prod_{l \neq v,w} (x + d'_l)\Big],$$

where $d = \operatorname{diag} N_e$, $d' = \operatorname{diag} N_f$, using $d_v = d'_v$ and $d_w = d'_w$, both of which hold because $v \notin f$ and $w \notin e$. The two products are monic of degree $n-2$, and $\sum_l d_l = \operatorname{tr} N_e = (b-1) + \operatorname{tr} M_S = \operatorname{tr} N_f = \sum_l d'_l$, so they agree in their coefficients of x^{n-2} and x^{n-3}. Their difference therefore has degree at most $n-4$, and so does Δ. $\square$

The bound is attained: the θ^2 part of μ has degree exactly $n-4$ for the Fano family, where it is $2y(y-1)(y-4)$, and for $\mathrm{PG}(2,3)$, where it has degree 9 (`code/curvature.py`). The step of the proof carrying the exponent, that two monic polynomials with equal root sums differ in degree at most $n-4$, is machine-checked as `CoefficientRigidity.prod_diff_natDegree_le`.

The second-order behaviour can be computed exactly rather than sampled, along curves that stay on the variety identically. Take hyperedges e, f and coordinates $v \in e \setminus f$, $w \in f \setminus e$, let $R(\theta)$ rotate the plane $\operatorname{span}\{\varepsilon_v, \varepsilon_w\}$, and apply it to P_e and P_f alone. Since $P_v + P_w$ is the identity on that plane, $R(P_e + P_f)R^\mathsf{T} = P_e + P_f$ identically in θ, so tightness is preserved exactly and not merely to leading order, while each rotated block stays an orthogonal projection of rank b and the family genuinely decommutes. For the Fano family the mixed characteristic polynomial is then computed in exact rational arithmetic, its odd part vanishes as the symmetry $\theta \mapsto -\theta$ requires, and

$$\lambda_{\max}(\theta) = y_0 - C\theta^2 + O(\theta^4), \qquad y_0 = 7.19605818500856\ldots, \quad C = 0.03717851401295\ldots,$$

with y_0 a root of $y^7 - 21y^6 + 168y^5 - 644y^4 + 1218y^3 - 1050y^2 + 336y - 24$ and C algebraic of the same degree. All 168 curves of this shape give the same C, as the 2-transitivity of the Fano plane requires.

Running the same computation on two further commuting families shows what C depends on, and it is not what one would guess.

family	(a,b)	n	y_0	C, $\lvert e \cap f\rvert = 1$	C, $\lvert e \cap f\rvert = 0$
Fano	$(3,3)$	7	7.196058185	0.037178514013	—
AG$(2,3)$	$(4,3)$	9	9.005835502	0.021354509344	0.041740312350
PG$(2,3)$	$(4,4)$	13	10.916978461	0.022914015843	—

There is no closed form $C(a,b)$, and not for want of data: AG$(2,3)$ carries two values at the single parameter pair $(4,3)$, according to whether the two hyperedges rotated are disjoint or meet. In the two projective planes every pair of lines meets, which is why each of them shows one value; the affine plane has parallel classes and shows both. So the curvature is a function of the pair of hyperedges, not of (a,b), and the quantity a scaling law would name does not exist. What is uniform is arithmetic rather than metric: in all three families μ_0 is irreducible of degree n and $\deg C = \deg y_0 = n$, so C generates the same field as the top root (`code/curvature.py`).

That is one diagonal entry of a second-order form, not definiteness, and the natural next step is unavailable for a reason worth recording. Linearising idempotency and tightness at the commuting point gives a space of dimension 63, spanned by differences $m^k_{ij} - m^l_{ij}$ over hyperedges containing exactly one of i,j. The variety does not fill it. Locating the nearest point of the variety to $A_0 + \varepsilon D$, the 42 directions whose two hyperedges lie on opposite sides of the pair — exactly the tangents of the curves above — sit at distance $O(\varepsilon^2)$ and are tangent, while the remaining 21 sit at distance $O(\varepsilon)$ and are not (`code/hessian.py`). The commuting point is therefore a *singular* point of the variety. Its tangent cone is moreover not a linear subspace: Q is quadratic, so it does not vanish on sums, and four of the ninety-one pairs of cross basis directions tested have $Q(D_1 + D_2) \neq 0$, reaching 2. A Hessian on the linearised space is therefore the wrong object. Local maximality therefore remains open, and what it needs is a second-order theory on that cone.

The same is true at AG$(2,3)$, where the linearised space has dimension 180 against a conjugation orbit of 36: the 131 cross directions give dist$/\varepsilon^2$ equal to 1.963 and 1.998 at $\varepsilon = 0.08$ and 0.02, and the 49 same-group directions give dist$/\varepsilon$ equal to 1.4128 and 1.4107. The exponents agree with the Fano family and only the constant differs.

Neither measurement is needed, as it turns out: the obstruction is algebraic and two blocks wide.

Proposition 19. *Let $\{P_k\}$ be a commuting tight family of rank-b projections with hyperedges $\{e_k\}$, and suppose there are e, f and vertices v, w with $w \in e \cap f$ and $v \notin e \cup f$. Then the* *direction D with $D_e = E_{vw} + E_{wv}$, $D_f = -(E_{vw} + E_{wv})$ and $D_k = 0$ otherwise lies in the kernel of the linearised constraints but is not tangent to the variety. Consequently the family is a singular point of it, and the tangent cone is not a linear subspace.*

Proof. Write $A_k = P_k + \varepsilon D_k + \varepsilon^2 X_k + O(\varepsilon^3)$. Idempotency at order two is $X_k - P_k X_k - X_k P_k = D_k^2$, whose (i,j) entry for a coordinate projection is $X_{ij}\big(1 - (P_k)_{ii} - (P_k)_{jj}\big)$. At a diagonal entry j the coefficient is $1 - 2(P_k)_{jj}$, which is -1 when $j \in e_k$ and $+1$ when $j \notin e_k$ and is never zero, so

$$(X_k)_{jj} = \sigma_k(j)\,(D_k^2)_{jj}, \qquad \sigma_k(j) = \begin{cases} -1, & j \in e_k, \\ +1, & j \notin e_k. \end{cases}$$

The diagonal of X_k is therefore determined; the freedom the equation leaves lies in the off-diagonal blocks, which never meet a diagonal entry. Now $D_e^2 = D_f^2 = E_{vv} + E_{ww}$, the sign squaring away, and $D_k^2 = 0$ for every other k. Since v lies in neither e nor f, $\sigma_e(v) = \sigma_f(v) = +1$, so

$$\sum_k (X_k)_{vv} = 1 + 1 + 0 + \cdots + 0 = 2,$$

contradicting $\sum_k X_k = 0$; at w, which lies in both, the same computation gives -2. Hence no X exists, so no curve on the variety has velocity D. That D lies in the kernel of the linearisation is the order-one identity $P_k D_k + D_k P_k = D_k$, immediate from $w \in e$, $v \notin e$ and the corresponding statement for f, together with $D_e + D_f = 0$. Finally the cross directions are tangent, by the explicit rotations above, so the cone meets the kernel in more than the origin while omitting D; and it is not closed under addition, since Q is quadratic and does not vanish on sums. □

The hypothesis asks only that some vertex lie in two hyperedges with a further vertex outside both, which holds in every commuting tight family with $a \geq 2$ and $n > |e \cup f|$; so the singularity is not a feature of the two families measured. For the cross configuration the same computation gives $\sigma_e(v) = -1$ and $\sigma_f(v) = +1$, which cancel, and that difference of sign is the whole difference between the two kinds. Both the local computation and the full linear system for X are checked in `code/singular.py`, the latter by comparing the rank of the coefficient matrix with that of the augmented matrix: 133 against 134 at Fano and 360 against 361 at AG$(2,3)$ for the same-group direction, with equality in both for the cross one. The scalar core is machine-checked as `TangentObstruction.no_second_order`.

That reduction is specific to projections, and the specificity is the point rather than a caveat. Relaxing idempotency destroys it immediately, since $A_k = (b/p)I_p$ is positive semidefinite of the right trace with $\sum_k A_k = aI$ and exceeds the band already at $p = 4$, as recorded in §7. Its rank is p rather than b, so what it shows is that trace and tightness do not suffice; the rank-n interpolation of §D is the statement that does keep the rank fixed. That is not evidence against Xu's conjecture, which is about projections; it is the obstruction recorded in §7 for $A_k = (b/p)I_p$, appearing again. It does mean any argument for the monotonicity must use idempotency and not only the rank, the trace and $\sum_k A_k = aI$, and it is why the search above enforces $\|A_k^2 - A_k\|$ rather than assuming it: without that check the retraction leaves the projection class silently, and the numbers it then produces are about the wrong class.

The identification in Proposition 10 is elementary and we make no claim of novelty for it. For diagonal matrices it is stated without proof by Marcus, Spielman and Srivastava [22, §7] in the square case $q = n$, with arbitrary diagonal weights; the statement here covers $q \neq n$, specialises the weights to 0/1 projections, supplies a proof, and carries the substitution $x = t^2$ with its degree bookkeeping. It should not be confused with the hypergraph Heilmann–Lieb theorem of Wan, Wang and Fan [4], which concerns a different polynomial — the matching polynomial of the adjacency hypermatrix, whose extreme root scales as $\Delta^{1/k}$ — and neither implies the other. Both routes of Proposition 10 are computed in `code/xu_sharp.py`, the mixed characteristic polynomial from the definition $\prod_e (1 - \partial_{z_e}) \det(xI + \sum_e z_e P_e)$ and the matching counts by subset dynamic programming, sharing no code and agreeing coefficient by coefficient in exact arithmetic on four hypergraphs at $b = 2, 3, 4$. The skeleton of the argument is machine-checked: the root identification of Proposition 10 as `XuSharp.bridge_roots`, the passage from a ceiling and a converging ladder to the supremum as `XuSharp.sharp_of_ladder`, and the consequence that no smaller constant can hold as `XuSharp.no_smaller_constant`. Godsil's bound and subgraph monotonicity enter there as hypotheses, being classical.

Four further files carry the material of this subsection and of §6.5, all with no `sorry` and on the three standard axioms. `RamaLean/SpectralNoGo.lean` is the obstruction of §7 in two halves that do not lean on each other: `const_of_factors_through_traces`, that any functional reading only the power traces of the blocks agrees on any two families of idempotents with equal block traces, which follows from $A^{j+1} = A$ and nothing else; and `spectral_bound_is_the_target`, that a bound both valid on the class and strong enough to give the target coincides with the target, with `target_gives_spectral_bound` for the converse. `RamaLean/ArcTangent.lean` carries the identity $\sum_j \sigma_k(j)(D_k^2)_{jj} = 0$ and the passage from a curve to Mathlib's Mathlib's own tangent cone. `RamaLean/OrderFourSolvable.lean` is the step joining `OrderFour.order_four_sum_zero` to `CokernelRank.cokernel_const`. `RamaLean/MixedDiscriminant.lean` carries the symmetry, diagonal value and additivity of the mixed discriminant. `RamaLean/ConeFix.lean` carries the entrywise solvability criterion behind Proposition 14 and the identification of the obstructed pairs as the twin pairs, and `RamaLean/ConeWitness.lean` the matrix level of the counterexample that makes the twin conditions necessary, every step by kernel computation. `RamaLean/OrderThreeLaw.lean` carries Lemma 15. `RamaLean/OrderThree.lean` is retained for its orbit lemmas; its `diag_mul_add_mul_of_cross` and `order_three_unobstructed` are true statements whose hypothesis no second-order correction satisfies, and nothing here cites them. `RamaLean/SpanRank.lean` carries the two scalar criteria behind the span and rank formulas, that a linear functional vanishes on $\{\sum t = 0\}$ exactly when its coefficients agree and that a diagonal quadratic form vanishes there exactly when every pair of coefficients cancels, with the pigeonhole that makes $|K(i,j)| \geq 3$ always an edge. `RamaLean/TwoRegular.lean` carries every step of Proposition 17 except the implicit function theorem itself: the divisibility, as the expansions of the quadratic and cubic pieces at the jet $tD + t^2 w$, where `quadratic_expand` is the one place $Q(D) = 0$ is consumed and is what removes the t^2 term; that an affine map with surjective linear part has a zero, which is the only place the rank hypothesis is used; and that a family of points of the set of the form $x_0 + tD + t^2 w$ with $t\,w \to 0$ puts D in the tangent cone, which is where the second-order shape does its work. What is cited and not formalised is stated in each file: the implicit function theorem step of

Proposition 17, and Aleksandrov's, Bapat's and Gurvits' theorems.

7 What a proof must contain

8 What a proof must contain

Each of the methods that does not reach the band fails for a reason that can be named exactly, and each such reason is a property that a proof must have. We collect them here as a specification rather than as a list of failures: a proof must not induct on vertex deletion, must not be a side-constant specialization of the multivariate matching polynomial, must control deficient vertices of the path tree, and must not pass through Floquet theory. Every one of these is sharp, in the sense that the method in question is exactly optimal for the class of objects it can see; that is the content of Section D.

The lower bound, unlike the upper bound, is not monotone under vertex deletion. The Heilmann–Lieb argument for $|x| \le 2\sqrt{\Delta-1}$ inducts on vertex deletion, which is available because Δ only decreases. The quantity $|\sqrt{a-1}-\sqrt{b-1}|$ has no such monotonicity: deleting one vertex of the larger side of $K_{3,4}$ produces $K_{3,3}$, which is regular, and the gap collapses to zero, for $a = b$ one has $c = 2\sqrt{m}$, so the interval $[c - 2\sqrt{m},\, c + 2\sqrt{m}]$ has its lower end at 0 and the statement is empty. So the naive induction by vertex deletion, which is what yields the upper bound, cannot prove the lower one.

That obstruction can be strengthened, and in a way that rules out the whole family of such arguments rather than one instance of it. Every b-regular bipartite graph is an induced subgraph of some (a, b)-biregular graph, by an explicit padding construction; and over b-regular bipartite graphs the least root of ν_G tends to 0, in closed form $2\sin(\pi/2N)$ for $b = 2$. So on the class obtained by closing the biregular graphs under vertex deletion, the infimum of the least root is 0, and the inequality to be proved is simply false there. Any argument that deletes vertices and applies an inductive hypothesis, cavity recursions, stability transfer, the Heilmann–Lieb induction, is an induction over that closure, and therefore cannot work, not because biregularity is lost but because the statement itself fails. An induction that carries something other than the target statement is not excluded, and §2.5 gives one: its hypothesis $q - r \ge 1$ is precisely a hypothesis the padding family above violates. For $\mathrm{AG}(2, 3)$, where $(s - t)^2 = 0.10102$, the minimum over all choices of d deleted vertices on the larger side runs 0.3593, 0.2424, 0.1383, 0.0549 for $d = 0, 1, 2, 3$: it drops below the target at depth three.

A second route, substituting side-constant values into the multivariate matching polynomial of Heilmann and Lieb, fails for a reason that is visible in one line. Writing $M_G(z; x) = \sum_M z^{|M|} \prod_{v \notin M} x_v$ and setting $x_v = \alpha$ on the p-side and $x_v = \beta$ on the q-side gives $\sum_k m_k z^k \alpha^{p-k} \beta^{q-k}$, in which the exponent pair contributes c^{p-q} under $(\alpha, \beta) \mapsto (c\alpha, \beta/c)$ independently of k; so the specialization depends on α and β only through the product $\alpha\beta$. Since the image of a product of two upper half-planes is $\mathbb{C} \setminus [0, \infty)$, stability under this substitution says exactly that the roots are real and nonnegative, and nothing more. In particular it says the same thing when $a = b$, where the conclusion to be proved is empty. No choice of α and β can see $a - b$.

Two obstructions remain in the direct approach, both caused by cycles in G. First, a leaf of the path tree that is an A-vertex carries $R_A = y > 0$, of the wrong sign. This one is disposed of: a maximal self-avoiding walk ending at $w \in A$ contains all a neighbours of w, hence at least a vertices of B and therefore at least a vertices of A, so

$$p < a \implies \text{the path tree has no } A\text{-leaf},$$

which covers every $K_{p,q}$ with $p < q$. Second, and this is what blocks the argument, an internal A-vertex of the path tree may have $k < P$ children, the walk having already consumed part of its neighbourhood; the bound $R_A \le y - P\alpha/(\alpha + Q)$ then fails. With $k = 1$ closure would require $\alpha(1-\alpha-Q)/(\alpha+Q) \ge y$, impossible for $b \ge 3$ since $Q \ge 2$. Deficiency occurs already in $K_{3,4}$, where the walk $a_1 b_1 a_2 b_2 a_3$ leaves a_3 with two children rather than three. What the argument as it stands establishes is that a *full* truncated biregular tree carries no eigenvalue in the gap, consistent with the deficient subtree P_8 carrying one; that is a statement about the tree and not about G. Controlling deficient A-vertices is the open problem.

The case $b \ge 3$ of the biregular family is the remaining open one, since $b = 2$ is Theorem 8; note that every graph named below has $a \ne b$, so the statement being tested is not the empty one. It has been tested at $d = 1$ on $K_{3,4}, K_{3,5}, K_{3,6}, K_{3,7}, K_{3,8}, K_{4,5}, K_{4,6}, K_{4,7}, K_{5,6}$ and a $(3,2)$ design incidence graph, with smallest-root ratios between 1.51 and 3.33 and no root in a gap.

The natural attack is Godsil's recursion $R_v = x - \sum_u 1/R_u$, whose fixed points on the (a,b)-biregular tree are real precisely off the spectrum. Asking that the region $R_b \in [x, P/x)$, $P = a-1$, $Q = b-1$, be closed under the recursion reduces to $u^2 - (2P+Q)u + P^2 > 0$ at $u = x^2$, i.e. to $u < \big(2P + Q - \sqrt{Q^2 + 4PQ}\big)/2$. This is satisfied throughout the gap with room to spare: the ratio of that threshold to $(\sqrt{P} - \sqrt{Q})^2$ ranges from 2.7 to 37.7 over the cases above. So the analytic part of an induction is not the difficulty.

What blocks it is the boundary. A leaf of the path tree carries $R = x > 0$, while the argument wants $R < 0$ at a-vertices, and mixed signs among the children of a b-vertex leave the sum uncontrolled. One might hope to exploit that the counterexample to the naive subtree argument, P_8 inside the $(3,2)$-biregular tree, is deficient at *internal* vertices, whereas a path tree is full at internal vertices; but that is false for graphs of small girth, where a self-avoiding walk can be blocked early, in K_4 the walk $0, 1, 2$ admits a single extension. We therefore leave $b \ge 3$ open.

A proof of Conjecture 1 cannot go through Floquet theory. For $b_1 = b$ the deck group of the universal cover is the free group F_b, and Floquet theory is exactly the statement that for an *abelian* deck group the fibres are indexed by the characters of that group; the two coincide only at $b = 1$, where $F_1 = \mathbb{Z}$. For $b \ge 2$ the correct index set is the unitary dual of F_b, and the relevant object is the adjacency operator in the reduced C^*-algebra of F_b. This is not merely a technical obstruction: the computations of [34] show that the per-component weight in the exponential formula, a single graph for $b = 1$, becomes a sum over the many non-isomorphic connected ℓ-covers for $b \ge 2$.

That framework is not empty. As $d \to \infty$, independent uniform random permutations are strongly asymptotically free [18], and consequently the spectrum of a random d-lift of G converges to $\mathrm{spec}(T)$. Conjecture 1 is a finite-d statement about the expected characteristic polynomial rather than an asymptotic statement about typical lifts, and its content is precisely that the containment holds uniformly in d, with no exceptional set. What the $b_1 = 1$ case shows is that the interval in [7] is not the natural receptacle for the roots, and that the correct one is spectral rather than metric.

9 The formalization, in full

A The formalization

The core algebraic derivation, the exponential formula, the closed form of Theorem 3 for all n, r (equivalently, everything from the cycle weights (1) to the closed form), its factorization, and the parity Corollary 7, is machine-checked in Lean 4 over Mathlib (v4.30). The main entry points:

- `Paper2.expFormula`, the exponential formula (3) in integer form $r\,A_r = \sum_k r^{(k)} f(k) A_{r-k}$ over any commutative ring, proved via Mathlib's Cauchy cycle-type count (`Equiv.Perm.card_of_cycleType_mul_eq`);
- `Paper2.thm1_Sr`, Theorem 3 in the form: the expected cycle weight (1) equals the closed form, for all n, r, over any $\mathbb{Q}$-algebra domain;
- `Paper2.cG_factored`, `Paper2.quotient_parity`, the factorization and Corollary 7.

The three remaining corollaries are now formalized in full. Corollary 4 is proved as a literal identity of formal power series, `Paper2GenFun.genU_mul` giving $(1 - 2YX + X^2)\sum_d U_d(Y)X^d = 1$ and `genPhi_mul` the companion form with numerator $(1 - X)^2$, by coefficient comparison against the Chebyshev recurrence. Corollary 5 is proved as the statement about roots, `Paper2Roots.Phi_ne_zero_of_two_lt_abs`: if $|x| > 2$ then $\Phi_{n,r}(x) \neq 0$, via $|T_n(x/2)| > 1$ (`one_lt_abs_cT`) and the vanishing of neither factor of the factorization. Corollary 6 is proved for all n and r, `Paper2FibLucas.cor2_phi3_general`, replacing the bounded `native_decide`; the route is the identity $F_{j+2m} + F_j = L_m F_{j+m}$ for even m (`fib_shift`), whose base cases are $F_{2m} = L_m F_m$ and, using Cassini, $F_{2m+1} + 1 = L_m F_{m+1}$, and which is stated without truncated subtraction by writing $L_m = 2F_{m+1} - F_m$.

Trust base for the general theorems: `propext`, `Classical.choice`, `Quot.sound` only (checked with `#print axioms`); the bounded `native_decide` checks additionally invoke `Lean.ofReduceBool`. No `sorry`. The single remaining unformalized ingredient of the *graph-theoretic* statement is the sentence before (1): a lift of a cycle is a disjoint union of cycles, elementary covering-space combinatorics for which Mathlib currently lacks the vocabulary.

For the biregular half, `RamaLean/BipartiteMatchingPoly.lean` supplies a matching polynomial for bipartite graphs, Mathlib having none, and formalizes: the deletion recursion $Q_G = Q_{G-v} - z\sum_{u\sim v} Q_{G-v-u}$ (`bipQ_delete_left`); the general Taylor shift $[X^j]f(X + c) = \sum_i \binom{i+j}{j} f_{i+j} c^i$ (`taylor_coeff_of_natDegree_le`), from which the coefficient formulas at $p-1$, $p-2$ and $p-4$ follow; the count $2m_2 + |\mathrm{confL}| + |\mathrm{confR}| = |E|(|E|-1)$ with the two conflict classes as degree sums; and the combinatorial inputs to Theorems 17 and 18, that three pairwise-meeting edges of a bipartite graph share a vertex (`three_meeting_share`), that the conflict degree of an edge is $(d_u - 1) + (d_v - 1)$ (`conflict_degree`), the trichotomy for four-cycles of the conflict graph (`four_cycle_trichotomy`), and the identity the whole expansion runs on, that the matching indicator is the alternating sum over subsets of the conflicting pairs (`alt_sum_conflictPairs`). Several of these, including the trichotomy, depend on no axioms at all. `RamaLean/Paper2Unicyclic.lean` further formalizes that the lower band edge is $(\sqrt{P} - \sqrt{Q})^2$ and so vanishes exactly when $P = Q$ (`gap_eq_zero_iff`), which is the triviality recorded above.

The inclusion–exclusion itself is formalized as a chain. The matching indicator is the alternating sum over subsets of the conflicting pairs of a set of edges (`alt_sum_conflictPairs`); substituting it turns m_k into a double sum (`mCount_eq_sum_alt`); because conflict is intrinsic to a pair of edges, the inner index set depends on the outer one only through a containment (`powerset_conflictPairs_eq`), so the two summations exchange (`mCount_inclusion_exclusion`); the resulting weight is a binomial coefficient, by a bijection between the k-subsets containing a fixed V and the $(k - |V|)$-subsets of the complement (`card_powersetCard_filter_superset`), giving

$$m_k = \sum_S (-1)^{|S|} \binom{|E| - |V(S)|}{k - |V(S)|}$$

over sets S of conflicting pairs (`mCount_closed_form`); and since the weight sees S only through $|S|$ and $|V(S)|$, the sum fibers over that pair (`mCount_grouped`, `mCount_grouped_range`), which is the grouping by subgraph type. The chain is exercised end to end at $k = 1$ (`mCount_one_via_inclusion_exclusion`): every nonempty set of conflicting pairs has at least two endpoints, so only the empty set survives the binomial and contributes $\binom{|E|}{1}$, reproving $m_1 = |E|$ through the inclusion–exclusion rather than by the direct bijection, so that two independent routes cross-check each link.

What is *not* formalized is the evaluation of the ten resulting counts as explicit functions of (p, a, b) and $C_4(G)$, the step at which `three_meeting_share` and `four_cycle_trichotomy` are applied, together with the numerical verifications. One obstacle there is worth recording: the formalized conflict set consists of *ordered* pairs, so each edge of the conflict graph occurs twice. This leaves the inclusion–exclusion untouched, since the alternating sum vanishes over any nonempty index set whether or not it double counts, but it means the grouping enumerates more configurations than the ten unordered types above, and aligning the two would require rebuilding the conflict set on unordered pairs.

The operator side of Sections [34]–D is formalized in four further files, all on the standard three axioms and with no `sorry`. `RamaLean/SignReflection.lean` carries Proposition 46 to the point where only the rank-one expansion of μ is left by hand: `sign_avg_reflect` proves that $\sum_{\varepsilon \in \{\pm 1\}^q} \det(tI - \sum_k \varepsilon_k S_k)$ is even or odd in t according to the parity of the dimension, with *no* hypothesis on the S_k, by the involution $\varepsilon \mapsto -\varepsilon$; `sum_prod_sgn` is character orthogonality on the Boolean cube, proved by the same involution applied to one coordinate; `sum_multiaffine` deduces that averaging a multi-affine function over signs returns 2^q times its constant term, which is exactly what $\prod_k (1 - \partial_{z_k})$ extracts; and `mixedChar_reflect` and `band_edges_swap` supply the shift and the identity $2a - (\sqrt{a-1} + 1)^2 = (\sqrt{a-1} - 1)^2$ that exchanges the band edges. `RamaLean/ProductBound.lean` proves Proposition 21 (`no_root_above`: a convex combination of products of positive numbers is positive) together with the exact overshoot $ab - (s+t)^2 = (\sqrt{m} - 1)^2$ (`edge_gap`) and its vanishing exactly at $m = 1$ (`edge_gap_eq_zero_iff`), which is the case $a = b = 2$ where the band is proved outright; it also proves Proposition 22, that at $b = 2$ the two band edges are the single condition $\pi_{\max} \le 4(a-1)/a^2$ (`band_iff_pi`), and that the strictly weaker $\pi_{\max} \le 4/a$ already yields $a + 2\sqrt{a}$, which beats the Marchenko–Pastur edge for every a and the product bound ab for $a > 4$ (`weak_bound`, `weak_beats_mp`, `weak_beats_ab`). `RamaLean/GramDet.lean` discharges the exterior-algebra inputs by writing them in Gram coordinates, where each becomes a determinant statement. `det_border` is the bordered determinant formula $\det\left[\begin{smallmatrix} a & r^\top \\ c & G \end{smallmatrix}\right] =$

$a \det G - r^{\mathsf{T}} \operatorname{adj}(G)c$, which is exactly (19); `det_gram_eq_zero_of_dep` is $f_k \wedge \omega'_k = 0$, the vanishing of a Gram determinant on a dependent family; and `hsimple_of_border_zero` combines them into $\|f_k\|^2 \|\omega'_k\|^2 = \|\iota_{f_k} \omega'_k\|^2$, which is precisely the hypothesis `hsimple` used below, so that hypothesis is now proved rather than assumed. `RamaLean/Tightness.lean` supplies the other input, `htight`: that $\operatorname{Adj}(A) = aI$ forces $\sum_k \iota_{f_k} \omega'_k = 0$. Working in coordinates, with a rank-two block presented by its two spanning vectors, the polarization step becomes three elementary facts, `iota_antisymm`, that $\langle a, \iota_b \omega\rangle = -\langle b, \iota_a \omega\rangle$, which is a two-term `ring` identity; `iota_proj_of_orthogonal`, that compressing the block to $e^\perp$ does not change the pairing against f_k; and that each $\iota_{f_k} \omega'_k$ already lies in $e^\perp$. `tight_sum_contraction_eq_zero` assembles them, and `crossTerm_nonneg_of_tight` composes the result with `CrossTerm.crossTerm_nonneg`, so that Theorem 33 is machine-checked from the tightness hypothesis alone. `RamaLean/Plucker.lean` closes the last link, the identification of `CrossTerm`'s coordinate vector `w` with the Plücker coordinates of ω'_k. Representing $b_1 \wedge b_2$ by its antisymmetric array on *ordered* pairs scaled by $1/\sqrt{2}$ makes `pl_dot`, that the resulting inner product is the Gram determinant, hold with no $i < j$ bookkeeping: the double count and the two factors of $1/\sqrt{2}$ cancel. `iota_eq_comb` writes $f_k = \iota_e \omega_k$ in the compressed plane with explicit coefficients $-\langle e, b_2\rangle$ and $\langle e, b_1\rangle$, and `hsimple_of_comb` then makes `hsimple` a `ring` identity rather than an assumption. `crossTerm_nonneg_plucker` assembles everything: **for a tight family of rank-two blocks and any unit vector, $C_2 \geq 0$ is machine-checked with no hypothesis remaining**.

The same holds at every level. `GramDet.border_gram` identifies the bordered Gram matrix of $[v \mid M]$ with the Gram matrix of the bordered family; `border_det_eq_zero_of_mem_span` then gives $f \wedge \omega = 0$ whenever f lies in the column span, which it always does; and `hsimple_of_mem_span` combines them into $\|f\|^2\|\omega\|^2 = \|\iota_f \omega\|^2$ for a wedge of *any* number of vectors. So the hypothesis `hsimple` of Theorem 34 is discharged at every r, not only at $r = 2$, and what Theorem 34 asserts, that each S-summand of C_r is a Schur square minus the square of a contraction, is machine-checked unconditionally. Finally `gram_det_basis_of_injective` and `gram_det_basis_of_not_injective` prove that for a coordinate family the Gram determinant of a set of blocks is 1 on matchings and 0 otherwise, that is $F_A = \mu_G$; that is the first of the two inputs to Proposition 35, and the second, the Kesten–McKay limit, is the only analytic ingredient anywhere in this section and is cited rather than proved. `RamaLean/CrossTerm.lean` carries Theorems 33 and 34. `crossTerm_eq_sq_sub` is the general level: with no hypothesis beyond `hsimple` it proves $\sum_{k \neq l} [\langle f_k, f_l\rangle \langle w_k, w_l\rangle - \langle u_k, u_l\rangle] = \|\sum_k f_k \otimes w_k\|^2 - \|\sum_k u_k\|^2$, which is the S-summand of Theorem 34 once $w_k = \omega'_k \wedge \omega'_S$ and $u_k = \iota_{f_k}(\omega'_k \wedge \omega'_S)$. Its two geometric inputs are hypotheses, `hsimple`, that $f_k \wedge \omega'_k = 0$ in its metric form $\|f_k\|^2\|\omega'_k\|^2 = \|u_k\|^2$, and `htight`, that $\sum_k u_k = 0$, and `crossTerm_eq_sq` specializes it to $r = 2$, where tightness kills the second square, and proves that they force C_2 to equal $\sum_{i,j} \big(\sum_k (f_k)_i (\omega'_k)_j\big)^2$, with `crossTerm_nonneg` and `crossTerm_eq_zero_iff` the consequences. The supporting `sum_gram_prod_eq_sum_sq` is the Schur product theorem in the case needed, that $\sum_{k,l} \langle f_k, f_l\rangle \langle w_k, w_l\rangle$ is a sum of squares. No exterior algebra is involved: the bivectors enter only through their coordinates. `RamaLean/VertexSplit.lean` carries the Gram form of Theorem 31. Its core, `det_add_vecMulVec`, is the adjugate form (18) of the matrix determinant lemma; Mathlib has the lemma only as $\det(A + uv^{\mathsf{T}}) = \det A \cdot \det(1 + v^{\mathsf{T}} A^{-1} u)$ and records the adjugate form as an explicit `TODO`, so this supplies it, under the hypothesis that $\det A$ is a unit, which is what the application needs, the Gram matrix of an independent family being positive definite. `det_le_det_add_vecMulVec` is the inequality the recursion consumes, $M_r(A) \geq M_r(A^{(e)})$, deduced from `AdjugatePSD.adjugate_posDef`. The singular case is scoped out as in `AdjugatePSD`: both sides of (18) are polynomial in the entries, so it extends by continuity from $A + \varepsilon I$, and that limit is not formalized. `RamaLean/CavityThreshold.lean` carries Proposition 32: `cavity_step` is the induction step, that child ratios at least $x/2$, nonnegative weights of total at most a, a nonpositive remainder and $4a \leq x^2$ force the parent ratio to be at least $x/2$ again; `threshold_iff` is its sharpness, that $x - 2a/x \geq x/2$ holds *if and only if* $x \geq 2\sqrt{a}$; `double_root_at_threshold` is the structural form of the same statement, the factorization $\rho^2 - 2\sqrt{a}\,\rho + a = (\rho - \sqrt{a})^2$ that collapses the interval of admissible thresholds to a point; and `no_slack_propagates` is Proposition 18; `cavity_step_deficient` is Proposition 42, the strengthened step; `compression_sum`, `trace_le_of_rayleigh` and `remainder_upper` are Proposition 43; and `ratio_form`, `one_add_le_inv_one_sub`, `inv_one_sub_sub_one_add`, `km_slack_calc` and `km_satisfies_target` are (22) and Proposition 44. Proposition 41 is `Tightness.adj_compressed`, with the per-block splitting `Tightness.iota_split`. `RamaLean/KestenMcKay.lean` carries Proposition 35: `radical_at_edge` that the radical is exactly 2 at the threshold, `G_at_edge` the closed form there, `inv_G_div_edge` the constant $1 + 1/(1 + \sqrt{a})$, and `excess_eq`, `excess_pos`, `excess_lt` that the excess is exactly $1/(1 + \sqrt{a})$, positive for every $a > 1$ and below any ε once $\sqrt{a} > 1/\varepsilon$. The two inputs it does not formalize are the ones that are not arithmetic: that F_A is the matching polynomial, which is Theorem 23. The Kesten–McKay step is no longer cited as a limit: `RamaLean/MomentTransfer.lean` carries Proposition 37 in full, namely the exact geometric expansion with remainder (`geom_remainder`), the remainder bound (`remainder_le`), the moment transfer (`stieltjes_close`), its form for reciprocals (`inv_close`) and the positivity of the gap between the support and the evaluation point (`edge_gap_pos`). What remains cited is the locality of root moments, which is [24] after [25], and Godsil's tree-like walk formulation [26]. The vertex recursion of Theorem 31 and the sign of X_e enter as hypotheses; what is machine-checked is that they close the induction, and that they close it exactly at $2\sqrt{a}$. `RamaLean/TracePSD.lean` and `RamaLean/AdjugatePSD.lean` carry the two positivity facts the extremal computation and the kernel recursion run on. In both, the standard input is a visible hypothesis rather than a hidden one: `trace_sq_le_sq_trace` takes the spectral decomposition of a positive semidefinite matrix as an explicit assumption, and `adjugate_posDef` is proved in the definite case, the semidefinite limit $A + \varepsilon I$ being left by hand.

`RamaLean/LaguerreBand.lean` carries Theorem 9 in full on the inequality side: `sqrt_amgm` and `sqrt_step` are the arithmetic–geometric mean step (4), `rowRadius_le` and `lastRow_le` are the two Gershgorin row bounds, `band_endpoints` identifies the disc with the biregular tree band, and `laguerre_band` assembles them through Mathlib's `eigenvalue_mem_ball`. What is not formalized is the classical passage from $\mu_{K_{p,q}}$ to the Laguerre polynomial and thence to its Jacobi matrix; that identification is a coefficient comparison and is done by hand above.

The three developments of [34] are formalized to the extent stated there. `RamaLean/AbelianCover.lean` proves that the integrand is real (`det_sub_smul_one_im`), that a continuous real function with vanishing average on a connected space has a zero (`exists_zero_of_integral_zero`), and the resulting localization and reduction; the Godsil–Gutman expansion is assumed as the hypothesis `haverage`.

`RamaLean/TreeSubstitution.lean` proves the substitution identity as a homogenization (`homogenize`), fixes its orientation on the star (`star_ratio`), and transports Heilmann–Lieb through a subdivision (`subdivision_root`, `subdivision_abs`); the covering combinatorics is by hand. `RamaLean/FeedbackVertex.lean` proves the combinatorial core (`no_bypass`, via Mathlib's characterization of acyclicity by path uniqueness), the resolvent diagonal, the algebraic collapse of the trace formula, and the trace step; the Pimsner–Voiculescu input appears as an explicit hypothesis of `feedback_one_of`, since it is not in Mathlib. Every theorem in all four files depends only on the three standard axioms, and none uses `native_decide`.

Remark 20. Everything here is stated for the uniform permutation model; by [7] the quotient coincides with the $(r-1)$-matching polynomial, so Theorem 3 can alternatively be deduced from [8], with the caveat about that preprint's status recorded in [34]'s preamble. We believe the generating function of Corollary 4, the evaluation of Corollary 6, and the formalization are new.

B Saturation, the path tree, and the free-convolution reading

B.1 The bound is per-vertex, and the saturation is only of constants

The limit at $(3, 6, 5)$ is real but narrower than it looks. What is saturated there is the *uniform* certificate: both bounds are attained with ratio 1.0000, so no choice of constants does better (`saturation_blocks_improvement`). The refinement below is not a choice of constants, and it escapes for that reason.

The requirement (9) caps every child by the same κ, evaluated at the worst child count $d-1$. But Proposition 13(i) already gives a left-type vertex with j children the sharper bound $\kappa(j) = \lambda + j/m$, and a right-type vertex with very few children sits where the path tree is nearly exhausted, so its children are child-poor too and carry a much smaller κ. Instantiating the bound at each child's own count rather than globally, the requirement at a right-type vertex becomes

$$k \;>\; \lambda\,\kappa(j_{\max}), \qquad j_{\max} = \max_i j_i,$$

which is strictly weaker whenever $j_{\max} < d - 1$ (`RatioRoute.refined_weaker`, `right_step_refined`). Measured on the path trees, it fails nowhere, including at the two $(3, 6, 5)$ points where the uniform requirement fails at eight vertices (`code/twolevel.py`). The saturation result is untouched: it says the constants cannot be improved, and this does not improve them.

Substituting Lemma 14 for $j_{\max}$ and asking the requirement for every admissible k removes the path tree altogether. With $\Delta = q - r$ it reduces, by one cancellation, to two inequalities:

Theorem 21. *If $\lambda < m$ and $\lambda^2 < \Delta$, then $\lambda\,\kappa(j) < k$ for every $k \geq \Delta$ and every $j \leq k - \Delta$.*

Proof. $\Delta(m-\lambda) > \lambda(\lambda m - \Delta)$ is equivalent to $\Delta > \lambda^2$ after cancelling m; since $m > \lambda$ and $k \geq \Delta$ the left side only grows with k (`RatioRoute.structural_closure`). □

Neither hypothesis mentions a path tree, a family or a graph. The second is the binding case already noted, where the children are leaves of ratio exactly λ; the first is what the per-vertex bound buys. Over 354 parameter points (d, q, r) that are *realizable* — a (d, q)-biregular graph with r vertices of degree q needs qr/d to be a positive integer — the pair holds at 326, or 92.1%, failing only as λ approaches the gap edge (`code/universal_close.py`). The elementary sharpening $k_u \le \min(d-1, k-\Delta)$, using that a child lies in L and has degree d, gains nothing at any of them (`code/sharp_close.py`): the obstruction is the two conditions themselves and not the counting.

B.2 The path tree inherits the gap

That obstruction is the invariant's and not the method's, and the reason is worth recording because it says the route is not exhausted. The recursion computes $F_v = \mu_{T_v}/\mu_{T_v - v}$, so "no F vanishes" is exactly $\mu_P(\lambda) \neq 0$ — strictly stronger than $\mu_G(\lambda) \neq 0$, since μ_P carries its own roots as well as those of μ_G. Were μ_P to vanish somewhere in the gap, no certificate of this kind could succeed there, whatever invariant it used.

It does not. On a tree $\mu_P = \prod_v F_v$ and the F_v are the pivots of the LDL factorization of $\lambda I - A_P$, so by Sylvester's law the number of negative F_v counts the eigenvalues of P above λ; comparing the count at the two ends of the gap counts the eigenvalues between, in time linear in $|P|$ and with no matrix. Over four (d, q, r) families that are *not* complete bipartite, with path trees of 3457, 17241, 22402 and 291429 vertices, the count is equal at both ends in every case: the path tree has no eigenvalue in the gap (`code/pathtree_inertia.py`). This is 'HEURISTIC', four families, and it is stronger than Godsil's divisibility, which places the roots of μ_G among those of μ_P and says nothing about where the others go.

So the statement the route proves is true throughout the gap, including where the sign-alternating certificate cannot hold. An argument therefore exists in principle at every λ in the gap. What does not exist is a better certificate of *this* family, and that is worth separating from the first point.

Two things settle it. Measuring $\min_v |F_v|$ across the gap suggests the sign-free alternative $|F_v| \ge \lambda$: the minimum equals λ exactly, attained at a leaf, at every fraction tested in two of three families, falling to 0.83λ and 0.51λ only at $(3, 6, 5)$ near the edge (`code/minF_scan.py`). But it is the harder condition, not a weaker one. With every child positive, $F_v = \lambda - \sum_u 1/F_u$ falls below λ automatically, and $|F_v| \ge \lambda$ additionally forces the sum past 2λ, which at a right-type vertex is $k \gtrsim 2\lambda B$ against the $k > \lambda B$ the sign condition needs — a factor of two in the child count.

And there is no cancellation to repair. The path tree is bipartite by depth, so the children of a vertex all lie on one side and are of one type, and same-type siblings never disagree in sign: over three families at six gap fractions each, including 0.99, the number of vertices whose children have mixed signs is *zero* in every case, out of 865, 4185 and 3220 parents (`code/signmix.py`).

B.3 The same measurement reformulates the conjecture

The inheritance of §B.2 says more than that the ratio route's target is true. It turns the conjecture into a statement about a finite tree.

Godsil gives $\mu_G \mid \mu_P$, and P is a tree, where the matching polynomial *is* the characteristic polynomial. So every root of μ_G is an eigenvalue of P, and

Proposition 22. $\mathrm{spec}(P) \subseteq \mathrm{spec}(T)$ *implies* $\mathrm{Zeros}(\mu_G) \subseteq \mathrm{spec}(T)$. *Conversely a root outside* $\mathrm{spec}(T)$ *is an eigenvalue of* P *outside it, so the containment fails exactly where the conjecture does.*

Proof. Immediate from $\mu_G \mid \mu_P$ in both directions (`PathTreeRoute.conj_of_pathtree`, `pathtree_fails_of_violation`). $\square$

The hypothesis is strictly stronger than the conclusion, since μ_P carries its own roots as well, and that is the point rather than a defect: it is a statement about the spectrum of one explicit finite tree with no matchings in it. It is also not vacuous in either direction. It fails for Hall's graph, necessarily — $\sqrt{5}$ is a root of μ_G, hence an eigenvalue of P, and the ratio system at $\sqrt{5}$ has decay 0.9636233789, so it is outside $\operatorname{spec}(T)$ — and a hypothesis provable in general would be worthless, the conjecture being false. And it holds wherever it has been measured, over the four families of §B.2.

The strictness of the hypothesis is decisive, and against it. At minimum degree three the containment is *false*. Over 86 graphs with $\delta \geq 3$ carrying a gap of $\operatorname{spec}(T)$, all but one inherit it; the exception is a 12-vertex graph with degrees $\{3, 4, 10\}$ whose path tree, on 3944 vertices, carries twelve eigenvalues inside the gap $[2.060, 2.160]$ while no root of μ_G lies there — counted by Sylvester's law, with the near-zero-pivot flag honoured, so the count is trustworthy (`code/pathtree_delta3.py`).

So $\operatorname{spec}(P) \subseteq \operatorname{spec}(T)$ fails exactly where Conjecture 1 holds, and Proposition 22 cannot carry a proof of the minimum-degree conjecture: no argument can pass through a hypothesis that is false on the graphs in question. What survives is the biregular reading. There the containment is measured to hold (§B.2), and there the proposition does reduce the inner end to a statement about one explicit finite tree. We record the negative result because it closes a route that looks plausible from the biregular evidence alone, and because it was the only reformulation five searches produced.

So the sign-alternating certificate is the best of its family, and its reach is exactly $\lambda^2 < \Delta$ — the binding case, where the children at the minimum count are leaves of ratio λ and the parent is $\lambda - \Delta/\lambda$. **That boundary is sharp rather than an artefact of the certificate's shape**, and extending the route past it needs an argument that is not an interval invariant on the ratios at all, since the target it would prove is true there and every certificate of this kind provably is not.

Finally, the sign separation is a consequence of bipartiteness: on wheels no natural class determines the sign of a path-tree ratio, endpoint degree, depth parity and child count each being mixed (`code/d3ratio.py`), so the certificate does not transfer to Conjecture 2 in general.

The biregular case is what survives, and §2.5 settles it on a nontrivial class by the ratio recursion. This part develops a second route to the same statement, through counting, the coefficient ladder and the operator form, together with the obstructions each of them meets. It uses the framework of Part I throughout; nothing in Part I depends on it, except that §2.5 and [34] each cite one result proved here.

C The band is the support of a free convolution power

There is a single reason behind all of this, and it is worth stating because it accounts uniformly for the failure of every method tried. Let $\chi = (1 - \frac{1}{b})\delta_0 + \frac{1}{b}\delta_b$, and let ν be the centred slot spectrum $\frac{1}{b}\delta_{b-1} + (1 - \frac{1}{b})\delta_{-1}$, so that $\chi = \delta_1 \boxplus \nu$.

Proposition 23. $\operatorname{supp}\big(\chi^{\boxplus a}\big) = \big[(\sqrt{a-1} - \sqrt{b-1})^2,\ (\sqrt{a-1} + \sqrt{b-1})^2\big]$.

Proof. χ is the spectral distribution of a rank-b projection scaled so that a free copies sum to the identity, and the free additive convolution power of such a measure is the spectral measure of the (a, b)-biregular tree, whose spectral radius is $\sqrt{a-1} + \sqrt{b-1}$ and whose spectrum is the stated interval; this is the same tree computation used in the proof of Corollary 11. It is classical and is used here only to name the band; no novelty is claimed, and nothing below depends on it beyond the identification of the two endpoints already established in Proposition 10 and Corollary 11. $\square$

So the band is the support of a free convolution *power*. Every method that fails computes instead a compound free Poisson of rate a with jump ν, whose free cumulants are $a\, m_n(\nu)$ rather than $a\, \kappa_n(\nu)$, and

$$m_n(\nu) = \kappa_n(\nu) \quad (n \leq 3), \qquad \kappa_4(\nu) - m_4(\nu) = -2(b-1)^2.$$

The two sequences agree through third order and part company at the fourth, by exactly $-2a(b-1)^2$. Order four is where every structure in this problem turns over, and it does so three independent times: the coefficients $E_0, \dots, E_3$ are determined by the parameters while E_4 carries the 4-cycle count; the moment and free-cumulant sequences of the jump agree up to order three and split at order four; and the parallel-class construction for the finite free convolution reproduces the elementary symmetric functions e_1, e_2, e_3 of μ and first fails at e_4. That is why the barrier bound, the scalar free convolution and the finite free convolution all overshoot by the same margin of about 1.1, and it is the analytic form of the third-moment obstruction above.

The finite free cumulants of μ itself follow the free power, not the Poisson:

$$\kappa_1 = a, \qquad \kappa_2 = \frac{ap(b-1)}{p-1}, \qquad \kappa_3 = \frac{ap^2(b-1)(b-2)}{(p-1)(p-2)},$$

with κ_4 the first that depends on the family, matching the coefficient ladder, where $E_0, \dots, E_3$ are universal and E_4 carries the 4-cycle count. Note $\kappa_3 = 0$ exactly when $b = 2$: Proposition 46 is visible in the cumulants.

Two further facts delimit what can be done with this. Writing $\psi_0(x) = x^{p-p/b}(x-b)^{p/b}$ and $\Psi = \psi_0^{\boxplus_p a}$, the polynomial Ψ satisfies the band at every p, but $\mu \neq \Psi$ in general and there is no real-rooted ρ with $\mu = \Psi \boxplus_p \rho$ unless the two agree: finite free deconvolution is unique, so such a ρ would have $\kappa_1 = \kappa_2 = 0$, and κ_2 is a positive multiple of the variance of the roots, forcing $\rho = x^p$. The very agreement of the first three cumulants is what forbids the factorization. Divisibility by a single copy, $\mu = \psi_0 \boxplus_p \rho$ with ρ real-rooted, would by the Marcus–Spielman–Srivastava bound give a lower bound on $\lambda_{\min}$, and it holds in small cases; but it is not a property of the class. Taking $b = 2$ and the incidence families of 3-regular graphs, the deconvolution ρ is real-rooted for $S(K_4)$ at $p = 4$ and for $K_{3,3}$ at $p = 6$, and fails for the cube Q_3 at $p = 8$ and the Petersen graph at $p = 10$, where the largest imaginary part of a root of ρ is 0.133 and 0.290 respectively. The cube is the informative case: it is 3-regular and bipartite, hence 1-factorable, so the failure is not caused by the absence of a partition of the blocks into parallel classes. Whatever separates the projection class from the positive semidefinite relaxation, single-copy divisibility is not it: at $(p, a, b) = (12, 4, 2)$ the scalar family, which violates the band, admits the factorization while a projection family does not.

Two general facts about free convolution came out of the attempt, and they bound what any argument of this shape can achieve. For finitely supported probability measures,

$$\min\operatorname{supp}\omega + \min\operatorname{supp}\tau \;\le\; \min\operatorname{supp}(\omega \boxplus \tau) \;\le\; \min\operatorname{supp}\tau + m_1(\omega), \tag{2}$$

the lower bound from the operator model and the upper from $\min\operatorname{supp}(\omega \boxplus \tau) = \sup_{w<0}[K_\tau(w) + R_\omega(w)]$ together with the monotonicity of R_ω, which follows from Cauchy–Schwarz. And if $\min\operatorname{supp}\tau \ge c$ then $\tau(\{c\}) \le \mu_2/\big(\mu_2+(m_1-c)^2\big)$, so by Bercovici–Voiculescu an atom of mass exceeding $\omega(\{\min\operatorname{supp}\omega\})$ pins the left edge of $\omega \boxplus \tau$ at c exactly.

Applied with $\omega = \chi$, for which $m_1 = 1$, (2) says that convolving with χ moves the left edge by at most 1. Any certificate that tries to reach the band edge by exhibiting μ as a free convolution with χ is therefore capped, and the first three cumulants do not constrain the other factor enough: there are measures with exactly the forced values of $\kappa_1, \kappa_2, \kappa_3$, real-rooted, supported in $[0, ab]$, whose convolution with χ has left edge 0. At $(a,b) = (5,4)$, $p = 12$ one may take $x^4(x-4)^3(x-6)^4(x-12)$, whose mass at 0 exceeds $1/b$.

The same atom mechanism explains the hypothesis $a \ge b$: for $a < b$ the measure $\chi^{\boxplus a}$ carries an atom at 0 of mass $1 - a/b$, so its left edge is 0 and the band statement is empty.

D Rank monotonicity, and what it explains

What does separate them can be said exactly, in terms of the representation (13). For a general positive semidefinite family with $\sum_k A_k = aI$ the marginal of s_k has probability generating function $\prod_i\big(\frac{a-\alpha_i}{a} + \frac{\alpha_i}{a}z\big)$ over the eigenvalues α_i of A_k, a Poisson binomial; it is exactly $\mathrm{Binomial}(b, 1/a)$ if and only if every α_i lies in $\{0,1\}$, that is if and only if A_k is a rank-b projection. Among Poisson binomials of fixed mean b/a with n equal parameters the variance is $(b/a)(1-b/(na))$, increasing in n; so the projections, where $n = b$, are the *variance-minimizing* extreme point of the whole family. The scalar counterexample $A_k = (b/p)I$ sits at the opposite end, at the Poisson limit, with variance exceeding the projection value by exactly b/a^2.

This has a consequence that organizes the whole picture. Fix (a,b) and let the common rank n of the A_k vary, with $A_k = (b/n)Q_k$ for rank-n projections Q_k summing to $(an/b)I$, so that $\operatorname{tr} A_k = b$ and $\sum_k A_k = aI$ throughout. At $(a,b) = (4,2)$, $p = 6$, $q = 12$ the extreme roots move monotonically:

n	least root	greatest root
2	1.129	6.871
3	1.004	7.706
4	0.951	8.090
5	0.921	8.317
6	0.903	8.467

against the band $[0.536, 7.464]$ at $n = b = 2$ and the Marchenko–Pastur interval $[0.343, 11.657]$ approached as $n \to p$. So the conjectured band is the endpoint at $n = b$ of a family of intervals increasing in n, whose other endpoint is Marchenko–Pastur, and the Marchenko–Pastur value is attained. That is the correct reading of the obstructions recorded above: each of them is sharp for the class it sees, namely the one with n unrestricted, and the conjecture is the assertion that restricting to $n = b$ contracts the interval all the way to the tree band.

The condition rank $A_k \le b$ is therefore the condition that the occupancy of a block fluctuates as little as possible at its given mean, and the conjecture asserts that minimal marginal fluctuation propagates to the extreme confinement recorded above. That suggests the right statement is one about measures rather than matrices: for a random vector $(s_1, \dots, s_q)$ of nonnegative integers with $\sum_k s_k = p$ almost surely, each s_k exactly Binomial$(b, 1/a)$, and joint generating function homogeneous of degree p and real stable, are the roots of $\mathbb{E}\prod_k(y - as_k)$ confined to the band? Every projection family induces such a law, and the scalar family does not, failing the marginal condition.

That class is rigid, which is what makes the reformulation faithful rather than a weakening. The marginal condition is itself a rigidity statement: in the polarized picture it holds if and only if the b slots inside each block are exactly jointly independent Bernoulli$(1/a)$. Consequently, any *determinantal* law satisfying the two combinatorial conditions has a rank-p projection kernel whose diagonal $b \times b$ blocks are $\frac{1}{a}I_b$, that is, it is the Naimark kernel of a projection family; and any law given by independent balls in bins satisfying them comes from an (a, b)-biregular incidence matrix, that is, from a commuting family. So the two canonical constructions of stable laws produce nothing new. At $b = 2$ more is true: for $(p, q, a, b) = (4, 6, 3, 2)$ and $(3, 6, 4, 2)$ the affine hull of the stable laws in the class coincides with the affine hull of the projection families, of dimensions 54 and 14 against 78 and 38 for the two combinatorial conditions alone. The measure formulation is therefore not a relaxation at $b = 2$.

The two combinatorial conditions alone do not suffice, and the point at which they fail is sharp: over all laws satisfying them the largest root is confined to the band only for q below an explicit threshold, and it reaches the trivial value ab exactly at $q = a^b$, the smallest q at which the binomial profile $n_j = q\binom{b}{j}(a-1)^{b-j}a^{-b}$ is integral. Stability is what must supply the rest, and it cannot be seen by any argument that symmetrizes: the divisibility constraint that stability imposes, namely that $(w + a - 1)^{b-1}$ divides $\mathbb{E}\big[s_l\, w^{s_k}\big]$, becomes vacuous on exchangeable laws. Negative association alone is likewise insufficient, by an explicit permutation counterexample whose polynomial has a root at ab.

The size of the claim is also worth recording. Since E_1 and E_2 are universal, the root distribution of μ has mean a and variance $a(b-1)$ for every admissible family. Measured against that, the lower band edge sits between 1.35 and 1.92 standard deviations below the mean over the range $3 \le a \le 100$, tending to 2 as a grows. The conjecture is therefore a statement of extreme concentration: the least of p roots is confined to about two standard deviations, where an unstructured sample of that size would reach out to order $\sqrt{\log p}$.

D.1 The extremal problem in dimension four

Graphs are not always extremal in that class, however, and the smallest case can be settled exactly. This is no longer needed for the band, at $p = 4$, $b = 2$ the reflection gives $\mu(t + a) = t^4 - 2at^2 + \beta$, and real-rootedness alone forces $0 \le \beta \le a^2$, hence $t^2 \le 2a \le 4(a-1)$, but it identifies the extremal families. Take $b = 2$, $p = 4$, so $q = 2a$ and $\sum_k P_k = aI_4$. Writing $g_{ij} = \operatorname{tr}(P_iP_j)$ and $h_{ij} = \operatorname{tr}\big((P_iP_j)^2\big)$,

Proposition 24. *For rank-2 orthogonal projections $P_1, \dots, P_q$ on $\mathbb{R}^4$ with $\sum_k P_k = \frac{q}{2}I$,*

$$E_4 = \frac{q^4}{16} - \frac{q^3}{4} + \frac{q}{2} + \frac{1}{4}\sum_{i \ne j}\big(g_{ij}^2 - h_{ij}\big),$$

and every summand is nonnegative. Equality holds exactly when $\operatorname{rank}(P_iP_j) \le 1$ *for all* $i \neq j$.

Nonnegativity is immediate once the right object is named: for $i \neq j$ put $X = P_jP_iP_j \succeq 0$; then $\operatorname{tr}\big((P_iP_j)^m\big) = \operatorname{tr}(X^m)$ for every $m \ge 1$, so $g_{ij}^2 - h_{ij} = (\operatorname{tr} X)^2 - \operatorname{tr} X^2 = 2\sum_{r<s} \lambda_r\lambda_s \ge 0$. The last step is formalized in `RamaLean/TracePSD.lean`: `sq_sum_sub_sum_sq` gives the identity $\big(\sum f\big)^2 - \sum f^2 = \sum_i \sum_{j \neq i} f_i f_j$ over any finite index set and `sum_sq_le_sq_sum` the inequality under nonnegativity, with the spectral input carried as explicit hypotheses on the matrix form.

Identifying a 2-plane $U \subset \mathbb{R}^4$ with its unit simple bivector $\omega_U \in \Lambda^2\mathbb{R}^4 \cong \mathbb{R}^6$ gives $g_{ij}^2 - h_{ij} = 2\langle \omega_i, \omega_j\rangle^2$, so the Welch bound for q unit vectors in a 6-dimensional space yields

$$E_4 \;\ge\; \frac{q^4}{16} - \frac{q^3}{4} + \frac{q^2}{12},$$

with equality exactly when $\{\omega_k\}$ is a tight frame of *simple* bivectors. This gives 30, 400/3, 1150/3 and 876 at $q = 6, 8, 10, 12$, matching the values found numerically.

At $q = 6$ the bound reads $E_4 \ge 30$, attained by the incidence family of K_4; and there the tight-frame condition degenerates to "orthonormal basis", which is the one value of q at which it can be met by simple bivectors alone. That is why graphs are extremal at $q = 6$ and not at $q = 8$, where the minimum 400/3 lies strictly below the graph minimum 134: the coincidence is the degeneration, not a principle. Even at $q = 6$ the graphs are only a slice of the minimizers, the equality locus is 4-dimensional modulo $O(4)$, and taking the ω_k from the six icosahedral axes, with the two $\mathbb{R}^3$ components Galois conjugate over $\mathbb{Q}(\sqrt{5})$, produces a noncommuting family with all $g_{ij} = 4/5$ and $E_4 = 30$. A lower bound of the same shape is known in the random setting: Brito, Dumitriu and Harris [12] show that a random bipartite biregular graph satisfies $\eta_{\min}^+ \ge \sqrt{d_1 - 1} - \sqrt{d_2 - 1} - \varepsilon_n$ with high probability. That is an asymptotic statement about adjacency eigenvalues of random graphs; Conjecture 1 is a deterministic statement about matching polynomial roots of every graph.

References

[1] A. D. Aleksandrov, *On the theory of mixed volumes of convex bodies IV. Mixed discriminants and mixed volumes*, Mat. Sb. (N.S.) **3** (1938), 227–251.

[2] R. B. Bapat, *Mixed discriminants of positive semidefinite matrices*, Linear Algebra Appl. **126** (1989), 107–124.

[3] L. Gurvits, *The van der Waerden conjecture for mixed discriminants*, Adv. Math. **200** (2006), 435–454.

[4] J.-C. Wan, Y. Wang, Y.-Z. Fan, *A hypergraph Heilmann–Lieb theorem*, arXiv:2206.09558 (2022).

[5] Z. Xu, Z. Xu, Z. Zhu, *Improved bounds in Weaver's* KS_r *conjecture for high rank positive semidefinite matrices*, arXiv:2106.09205 (2021).

[6] C. Godsil, I. Gutman, *On the matching polynomial of a graph*, in: Algebraic Methods in Graph Theory, 1981.

[7] C. Hall, D. Puder, W. Sawin, *Ramanujan coverings of graphs*, Adv. Math. **323** (2018), 367–410.

[8] G. Cochran, C. Groothuis, A. Herring, R. Rohatgi, E. Stucky, *A new (combinatorial) proof of the commutativity of matching polynomials for cycles*, arXiv:1810.05889 (2018). Preprint; no journal reference or DOI as of August 2026.

[9] J. Wan, W. Wang, A. Mohammadian, *On the matching polynomials of graphs with small number of cycles*, arXiv:2103.10580 (2021).

[10] Y.-M. Song, Y.-Z. Fan, Z. Miao, *Hypergraph coverings and Ramanujan hypergraphs*, arXiv:2310.01771 (2023).

[11] W. Yan, Y.-N. Yeh, *On the matching polynomial of subdivision graphs*, Discrete Appl. Math. **157** (2009) 195–200. The identity used here is their Corollary 2.1: for G d-regular, $m(S(G),x) = x^{m-n}m(G,x^2-d)$.

[12] G. Brito, I. Dumitriu, K. D. Harris, *Spectral gap in random bipartite biregular graphs and applications*, Combin. Probab. Comput. **31** (2022) 229–267.

[13] O. J. Heilmann, E. H. Lieb, *Theory of monomer-dimer systems*, Comm. Math. Phys. **25** (1972) 190–232.

[14] V. Y. Chernyak, M. Chertkov, *Fermions and loops on graphs. II. A monomer-dimer model as a series of determinants*, J. Stat. Mech. (2008) P12012.

[15] Y. Watanabe, M. Chertkov, *Belief propagation and loop calculus for the permanent of a non-negative matrix*, J. Phys. A **43** (2010) 242002.

[16] M. Jerrum, *Two-dimensional monomer-dimer systems are computationally intractable*, J. Statist. Phys. **48** (1987) 121–134.

[17] Y. Watanabe, K. Fukumizu, *New graph polynomials from the Bethe approximation of the Ising partition function*, Combin. Probab. Comput. **20** (2011) 299–320.

[18] C. Bordenave, B. Collins, *Eigenvalues of random lifts and polynomials of random permutation matrices*, Ann. of Math. **190** (2019) 811–875.

[19] J. P. McSorley, P. Feinsilver, *Multivariate matching polynomials of cyclically labelled graphs*, Discrete Math. **309** (2009) 3205–3218.

[20] H. Mizuno, I. Sato, *Characteristic polynomials of some graph coverings*, Discrete Math. **142** (1995) 295–298.

[21] C. Dalfó, M. A. Fiol, M. Miller, J. Ryan, J. Širáň, *An algebraic approach to lifts of digraphs*, Discrete Appl. Math. **269** (2019) 68–76.

[22] A. W. Marcus, D. A. Spielman, N. Srivastava, *Interlacing families II: mixed characteristic polynomials and the Kadison–Singer problem*, Ann. of Math. (2) **182** (2015), 327–350.

[23] A. Marcus, D. Spielman, N. Srivastava, *Interlacing families I: Bipartite Ramanujan graphs of all degrees*, Ann. of Math. **182** (2015), 307–325.

[24] P. Csikvári, P. E. Frenkel, *Benjamini–Schramm continuity of root moments of graph polynomials*, European J. Combin. **52** (2016), part B, 302–320.

[25] M. Abért, T. Hubai, *Benjamini–Schramm convergence and the distribution of chromatic roots for sparse graphs*, Combinatorica **35** (2015), 127–151.

[26] C. D. Godsil, *Matchings and walks in graphs*, J. Graph Theory **5** (1981), 285–297.

[27] C. Hall, *A simple bipartite counterexample to Conjecture 10: spectral localization of the ordinary matching polynomial*, verification note, personal communication, August 2026.

[28] E. R. Avakov, *Extremum conditions for smooth problems with equality-type constraints*, USSR Comput. Math. Math. Phys. **25** (1985), 24–32.

[29] A. V. Arutyunov, *Optimality Conditions: Abnormal and Degenerate Problems*, Kluwer, 2000.

[30] M. Ravichandran, J. Leake, *Mixed determinants and the Kadison–Singer problem*, Math. Ann. **377** (2020), 511–541.

[31] O. Angel, J. Friedman, S. Hoory, *The non-backtracking spectrum of the universal cover of a graph*, Trans. Amer. Math. Soc. **367** (2015), 4287–4318; arXiv:0712.0192.

[32] F. Bencs, P. Csikvári, *Evaluations of Tutte polynomials of regular graphs*, arXiv:2105.06798 (2021).

[33] V. Bonfioli, *The biregular case of a spectral-inclusion conjecture: the weighted plane class and its obstructions*, 2026.

[34] V. Bonfioli, *Roots of d-matching polynomials and the spectrum of the universal cover: a closed form at first Betti number one, and a refuted conjecture*, companion paper, 2026.